An Integrated Correction Method for Geometric, Atmospheric, and Terrain Effects in Hyperspectral Remote Sensing Images

Guorui Jia, *Senior Member, IEEE*, Yujie Zhao, Shuhan Liu, Chenchao Xiao, and Huijie Zhao

***Abstract*—Reflectance inversion is a critical challenge in quantitative hyperspectral remote sensing, especially in rugged terrain. The under-/over-correction of reflectance caused by such terrain and its impacts on subsequent information extraction have not been well quantified or effectively eliminated. To improve surface spectral reflectance inversion under varying illumination, atmospheric, and terrain conditions, an integrated correction method for geometric, atmospheric, and terrain effects (ICGAT) is proposed. The method incorporates a coupled radiative transfer model that accounts for direct solar irradiance, diffuse sky irradiance, surface-atmosphere coupling irradiance, and irradiance from neighboring slopes along the sun–terrain-target path. The geometric positioning and irradiance components are first computed, followed by an iterative retrieval of reflectance. Geometric resampling is subsequently applied to the reflectance, and an uncertainty quantification is finally conducted in the process. Tests show that ICGAT significantly improves spectral consistency between neighboring shaded and sunny slopes, reducing the influence of terrain relief on reflectance, as well as retrieving high-accuracy surface reflectance in flat areas. Meanwhile, the uncertainty propagation analysis could reveal not only the fidelity of the retrieved reflectance but also the major sources of uncertainty.**

***Index Terms*—Atmospheric correction, geometric correction, hyperspectral remote sensing, reflectance, topographic correction, uncertainty quantification.**

I. Introduction

In RECENT years, many hyperspectral remote sensing satellites/payloads have been launched worldwide. These include China's Advanced Hyperspectral Imagers (AHSIs) onboard Gaofen-5 (GF-5) [1], Gaofen-5B (GF-5B) [2], Ziyuan-1 02D (ZY-1 02D) [3], and Ziyuan-1 02E (ZY-1 02E) [4] satellites, the United States' Earth Surface Mineral Dust Source Investigation (EMIT) [5] on the International Space Station (ISS), Germany's Environmental Mapping and Analysis Program (EnMAP) [6] satellite and the DLR Earth Sensing Imaging Spectrometer (DESIS) [7] on the ISS, as well as Italy's Precursore Iperspettrale della Missione Applicativa (PRISMA) [8] satellite. There are also forthcoming instruments, including the Copernicus Hyperspectral Imaging Mission for the Environment (CHIME) [9] initiated by European Space Agency and the Hyperspectral Infrared Imager (HyspIRI) [10] by U.S. National Aeronautics and Space Administration.

To achieve subsequent quantitative applications, surface reflectance needs to be retrieved from the hyperspectral remote sensing images through the processing step of atmospheric correction. The traditional atmospheric correction algorithms are mainly based on the assumption of a flat-Lambertian surface, and can be categorized into three types: those based on the radiative transfer model, those belonging to the empirical linear method, and those that are a hybrid of the two [11]. The atmospheric correction codes based on the radiative transfer model include the atmospheric removal program (ATREM) [12], fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASHs) [13], atmospheric correction now (ACORN) [14], high-accuracy atmospheric correction for hyperspectral data (HATCH) [15], and so on. Among these, the FLAASH is widely used, considering the radiative components entering the sensor as composed of radiance from target reflection, background radiance, and atmospheric radiance. FLAASH utilizes the moderate resolution atmospheric transmission (MODTRAN) [16] model for radiative transfer calculations, and after an optional aerosol and columnar water vapor retrieval, it obtains the spectral reflectance of each pixel in the image. The empirical line correction is widely used in the airborne remote sensing community. And the hybrid methods use the field measurement to remove the residual errors commonly associated with sensor artifacts and radiative transfer models [11].

However, from the perspective of global resource exploration and surveys, rugged terrain will inevitably be involved. Hence, the impact of terrain on the reflectance accuracy cannot be ignored. The trend in land surface reflectance inversion is moving toward an integrated model of atmospheric and topographic correction.

To address the coupled influence of terrain and atmosphere on surface reflectance, several radiative transfer models have been developed that explicitly integrate topographic effects into the surface-atmosphere system [17], [18], [19], [20], [21].

Received 21 October 2025; revised 18 November 2025; accepted 16 December 2025. Date of publication 18 December 2025; date of current version 2 January 2026. *(Corresponding author: Huijie Zhao.)*

Guorui Jia and Yujie Zhao are with the Key Laboratory of Precision Opto-Mechatronics Technology, Ministry of Education, School of Instrumentation and Optoelectronic Engineering, Beihang University, Beijing 100191, China.

Shuhan Liu and Chenchao Xiao are with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources, Beijing 100048, China.

Huijie Zhao is with the Key Laboratory of Aerospace Broad Band Detection and Intelligent Perception, Ministry of Industry and Information Technology, School of Artificial Intelligence, Beihang University, Beijing 100191, China (e-mail: hjzhao@buaa.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3646001

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

These coupled models extend the conventional flat-Lambertian radiative transfer framework by incorporating local illumination geometry, shaded factor, slope and aspect variations, and adjacency effects from surrounding terrain. In recent years, a number of improved atmospheric–terrain coupled correction models have also been proposed [22], [23], [24], [25], aiming to better represent the complex radiative interactions over rugged surfaces. A comprehensive review of terrain correction methodologies can be found in [26], which provides a systematic summary of recent advances in this field.

Building upon these theoretical foundations, several operational algorithms have been developed for hyperspectral images. For nonhomogeneous and rugged surfaces, the relatively mature codes for hyperspectral images are Atmospheric Correction Software for Airborne and Satellite Imaging (ATCOR) [17], [27] and python-based atmospheric correction (PACO) [18]. ATCOR utilizes MODTRAN to perform radiative calculations and treats the radiation arriving at rugged surfaces as a combination of direct solar irradiance, diffuse sky irradiance, and surrounding terrain reflection irradiance, deriving terrain effects from surface elevation information in digital elevation model (DEM) data. However, it simplifies the calculation of radiation from nearby slopes and ignores multiple reflections between the surface and the atmosphere, resulting in under-correction or over-correction of the variation of light intensity and shadow effects on some sloping surfaces [27]. In the retrieval calculation, ATCOR uses a fixed value of 0.1 as the initial terrain reflectance and iterates three rounds to gradually eliminate the influence of the tilting surface and surrounding object [17]. PACO is based on the same theoretical basis as ATCOR and is the processor for EnMAP L2A products.

Recently, a few other atmospheric–terrain correction methods based on MODTRAN have been proposed. Utilizing MODTRAN 4, the work in [19] proposed an approach that differentiates between direct solar irradiance and diffuse sky irradiance components to simulate environmental effects on remote sensing signals while performing atmospheric and terrain corrections. However, this method does not account for the influence of the surrounding objects' reflected irradiance. In addition, it omits the impact of background objects when calculating the initial reflectance value and is limited to one or two rounds of reflectance computations. The absence of multiple iteration steps in reflectance retrieval may result in reduced accuracy. Other atmospheric–terrain correction methods [21], [28], do not incorporate iterative calculations. Specifically, [28] reported under-correction in certain shaded slope areas, while [21] adopted relatively coarse calculations of path radiance and sky-view factor.

Besides atmospheric–terrain correction, geometric correction is also necessarily conducted to remotely sensed images to remove geometric distortions induced by lens, atmosphere, and topographic relief, so that individual pixels have their proper planimetric map coordinates (x, y) [29]. Daniel proposed a parametric geocoding procedure called parametric geocoding (PARGE) [30] and successfully combined it with the radiometric and atmospheric correction package ATCOR [31]. This combination enables the sequential correction of geometric and radiometric aspects. However, the program required spatial resampling of the at-sensor radiance image prior to atmospheric–terrain correction, which inevitably affects the radiometric quantity. It would be improved if an integrated correction for atmospheric, geometry, and terrain effects had been achieved.

In addition, all the above methods adopt DEM data to depict the rugged surface. As a result, the accuracy of the DEM data used in the methods also affects the results of the final reflectance [32]. At present, the visual effect and the reduction of the correlation between reflectance and terrain factor are used to verify the effectiveness of the algorithm. This is because rugged areas make it difficult to obtain ground truth reflectance at the scale of remote sensing pixels.

Thompson et al. [33] introduced an optimal estimation approach to address the challenge of atmospheric correction. They simultaneously estimated atmospheric columnar water vapor, aerosol optical depth, and surface reflectance using a Bayesian estimation method, based on a prior library of surface spectral reflectance and atmospheric status. This approach yielded promising results, enabling the characterization of posterior uncertainty in both surface reflectance and atmospheric parameters through the estimated variance. By incorporating the influence of terrain correction (considering solar incidence angle cosine) into Thompson's precedent atmospheric correction model, Carmon et al. [34] proposed an integrated method also capable of terrain perception. Experimental research was carried out on Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) data, and the accuracy was evaluated through visual inspection of the shadow correction effect as well as reflectivity and terrain decorrelation. In addition, recent studies have explored the estimation of topographic parameters directly from radiance measurements [35]. In that, the relative consistency of the reflectance of the same material on a conical roof is improved, showing the effectiveness of the algorithm. However, this type of optimal estimation method essentially relies on a priori database, using an iterative optimization algorithm to ensure that the retrieved reflectance converges toward the prior values while the forward model's simulated radiance approaches the observed radiance. As a result, the retrieved reflectance will be influenced by the a priori reflectance, which is necessary to cover as many various ground objects' conditions as possible.

This article proposes an integrated correction method for geometric–atmospheric–terrain effects in hyperspectral remote sensing images. The model considers the coupling irradiance component between the rugged surface and the atmosphere in addition to direct solar irradiance, diffuse sky irradiance, and irradiance from neighboring slopes. The influences of both the target and the background are considered during the calculation of initial reflectance values. Subsequently, the final reflectance is solved through iterations. At the same time, an uncertainty estimate of the retrieved reflectance is generated according to the Guide to the Expression of Uncertainty in Measurement (GUM), which can be used for quality assessment of the surface reflectance in rugged areas.

Section II gives a detailed description of the integrated correction method for geometric, atmospheric, and terrain effects

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

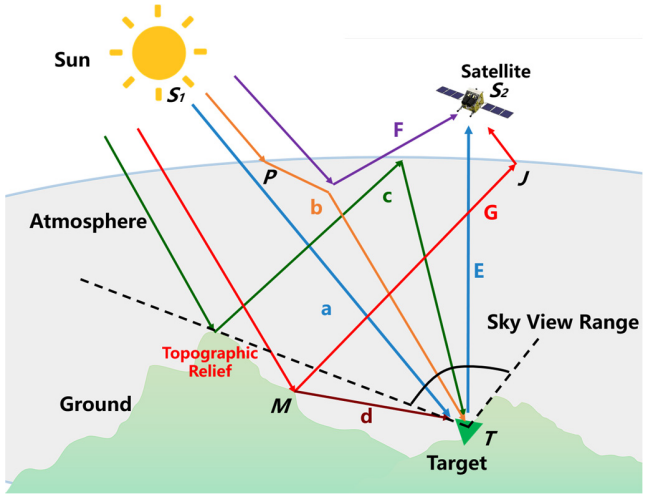

Fig. 1. Schematic of sun-target-sensor radiative transfer paths.

(ICGAT) method. Section III presents the description of the data used in the validation experiments and the corresponding results. In addition, a series of analyses is conducted on the experimental results, demonstrating the accuracy and the uncertainty characteristics of the rugged surface reflectance and illustrating its impact on remote sensing applications. Section IV discusses the main sources of uncertainty in reflectance data and analyzes the impact of terrain on reflectance retrieval. Section V draws conclusions.

II. Methods

To conduct the geometric–atmospheric–terrain correction for hyperspectral remote sensing data, it is necessary to determine the 3-D geometric radiative transfer path for each pixel first. Then, based on this, the radiative components above the rugged terrain that interacted with the pixel and finally reached the sensor could be solved. An integrated correction should be carried out to invert the reflectance image with only one spatial resampling as the final step to hold the geometric-radiative relationship as much as possible. The physical model, the iterative inversion of reflectance, and the quantification of reflectance uncertainty are described in turn as follows. In this study, the surface is assumed to be Lambertian, which is a common simplification in radiative transfer-based correction models. The implications of this assumption are further discussed in Section IV-C.

A. Physical Model

Modeling the radiative components first requires consideration of the radiative transfer process, which can be decided along the path of sun-target-sensor, as shown in Fig. 1.

Among them, four radiative components are in the sun-surface path. Direct solar irradiance (a in Fig. 1) is the direct solar beam transmitted through the atmosphere and onto the ground target. Diffuse sky irradiance (b in Fig. 1) is composed of the directional atmosphere-scattered solar radiation in the sky-view range of the target. Surface-atmosphere coupling irradiance (c in Fig. 1) represents the portion of energy that undergoes multiple interactions between the atmosphere and the rugged surface, experiencing backscattering and reflection before reaching the target [21], [36]. Irradiance from neighboring slope (d in Fig. 1) represents the portion of radiation received by the target pixel that is reflected from the surrounding rugged pixels. The terrain adjacency effect in the radiative transfer process refers to the surface-atmosphere coupling irradiance and the irradiance from neighboring slopes [37].

The surface-sensor path mainly consists of three components: target reflected radiance (E in Fig. 1), which comes from all the radiative components along the sun-surface path, reflected by the target surface, and passes through the atmosphere to reach the sensor entrance pupil; path radiance (F in Fig. 1), which is the upwelling scattered solar radiation by the atmosphere without interaction with ground; atmospheric adjacency effect (G in Fig. 1), which refers to the radiance reflected by untargeted pixels and scattered by the atmosphere.

In Fig. 1, S_1 is the location of the sun, P represents the scattering point of diffuse sky irradiance, T is the target pixel, M represents the adjacent pixel on a neighboring slope, J represents the scattering point of the atmospheric adjacency effect, and S_2 is the satellite observation position.

In general, the at-sensor radiance can be expressed as the sum of the target reflected radiance, path radiance, and atmospheric adjacency effect. The target reflected radiance comes from direct solar irradiance, diffuse sky irradiance, irradiance from neighboring slope, and surface-atmosphere coupling irradiance, as shown in the following equation:

$$\begin{aligned} L(x,y,z,\lambda) = {} & \frac{\rho_t}{\pi} \cdot T^{\uparrow}(z,\lambda) \cdot E^{*}_{\mathrm{dir}}(x,y,z,\lambda) \\ & + \frac{\rho_t}{\pi} \cdot T^{\uparrow}(z,\lambda) \cdot E^{*}_{\mathrm{dif}}(x,y,z,\lambda) \\ & + \frac{\rho_t}{\pi} \cdot T^{\uparrow}(z,\lambda) \cdot E^{*}_{\mathrm{ref}}(x,y,z,\lambda) \\ & + \frac{\rho_t}{\pi} \cdot T^{\uparrow}(z,\lambda) \cdot E^{*}_{\mathrm{coup}}(x,y,z,\lambda) \\ & + L_{\mathrm{atm}}(x,y,z,\lambda) + L_{a_\mathrm{adj}}(x,y,z,\lambda). \end{aligned} \tag{1}$$

In which $L(x,y,z,\lambda)$ represents the at-sensor spectral radiance corresponding to the ground target with 3-D coordinates (x,y,z), $E^{*}_{\mathrm{dir}}(x,y,z,\lambda)$ represents the direct solar irradiance, $E^{*}_{\mathrm{dif}}(x,y,z,\lambda)$ represents the diffuse sky irradiance, $E^{*}_{\mathrm{coup}}(x,y,z,\lambda)$ represents the surface-atmosphere coupling irradiance, $E^{*}_{\mathrm{ref}}(x,y,z,\lambda)$ represents irradiance from the neighboring slope, $L_{a_\mathrm{adj}}(x,y,z,\lambda)$ represents the atmospheric adjacency effect, and $L_{\mathrm{atm}}(x,y,z,\lambda)$ represents the path radiance, $T^{\uparrow}(z,\lambda)$ represents the atmospheric transmittance for upwelling radiation, ρ_t represents the reflectance of the target surface, and λ represents the wavelength.

The radiation components are calculated as follows.

The direct solar irradiance in rugged terrain is calculated, accounting for the surface orientation and shading effect caused by the topographic relief

$$\begin{aligned} E^{*}_{\mathrm{dir}}(x,y,z,\lambda) &= \left(\frac{r}{r_0}\right)^2 \cdot E_0(\lambda) \cdot T^{\downarrow}(z,\lambda) \cdot \Theta \cdot \frac{\cos\theta_i(x,y,z)}{\cos\theta_s(x,y)} \\ &= E_{\mathrm{dir}}(z,\lambda) \cdot \Theta \cdot \frac{\cos\theta_i(x,y,z)}{\cos\theta_s(x,y)} \end{aligned} \tag{2}$$

where r is the solar-Earth distance at the moment of imaging, r_0 is the average solar-terrestrial distance, $E_0(\lambda)$ is the top of atmospheric (TOA) solar irradiance, $T^{\downarrow}(z,\lambda)$ is the downward

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

transmittance, which is related to elevation z, θ_i is the solar incidence angle onto the surface, and θ_s is the solar zenith angle (SZA). $\Theta \in [0,1]$ is the shaded factor, which represents the shadow condition of the rugged terrain area at the position (x, y, z). And it is calculated according to the depth buffering algorithm [38]. E_{dir} is the direct solar irradiance on a horizontal surface.

By using latitude, longitude, and imaging time, the SZA θ_s and solar azimuth angle ϕ_s can be calculated. Subsequently, with the slope S and the aspect A, the cosine of the solar incidence angle for each pixel can be calculated [17]

$$\cos(\theta_i) = \cos S \cos\theta_s + \sin S \sin\theta_s \cos(A - \phi_s). \tag{3}$$

The diffuse sky irradiance on a sloped surface is the integration of the atmosphere-scattered radiance over the sky-view range of the target on the rugged terrain, and is composed of the anisotropic circumsolar (near solar direction) radiation and the isotropic sky radiation [17]

$$E^*_{\text{dif}}(x, y, z, \lambda) = E_{\text{dif}}\left[\Theta T^{\downarrow}\frac{\cos\theta_i}{\cos\theta_s} + \left(1 - \Theta T^{\downarrow}\right) V_{\text{sky}}\right] \tag{4}$$

where E_{dif} is the diffuse sky irradiance over a horizontal surface, V_{sky} is the sky-view factor, and $\Theta, T^{\downarrow}, \theta_i$, and θ_s are the same as in (2), respectively, omitting the corresponding independent variables for brevity hereinafter.

The sky-view factor is the ratio of the diffuse sky irradiance at the target to that on an unobstructed horizontal surface [39]

$$V_{\text{sky}} = \frac{1}{2\pi}\int_0^{2\pi} \times \left[\cos S \sin^2 H_\phi + \sin S \cos(\phi - A)\left(H_\phi - \sin H_\phi \cos H_\phi\right)\right] d\phi \tag{5}$$

where H_ϕ represents the unobstructed zenith angle range (i.e., from the zenith downward to the local horizon) at the target for each direction ϕ, with ϕ value ranging from 0° to 360°.

The surface-atmosphere coupling irradiance arises from the radiation bouncing between the atmosphere and the ground surface single or multiple times to reach the target. When the reflectance of the surrounding objects is unknown, it is assumed that the Earth-atmosphere coupling irradiance can be calculated through a summation of series based on the equivalent reflectance of the surrounding objects ρ_e and the atmospheric hemisphere albedo $s(\lambda)$, accounting for the obstruction of the sky by the rugged surface

$$E^*_{\text{coup}}(x, y, z, \lambda) = \bar{E}_{\text{coup}}(x, y, z, \lambda) \cdot V_{\text{sky}} \tag{6}$$

$$\bar{E}_{\text{coup}}(x, y, z, \lambda) = \frac{\left[E^*_{\text{dir}}(x, y, z, \lambda) + E^*_{\text{dif}}(x, y, z, \lambda)\right] \cdot \rho_e(\lambda) \cdot s(\lambda)}{1 - \rho_e(\lambda) \cdot s(\lambda)} \tag{7}$$

where $\bar{E}_{\text{coup}}$ represents the surface-atmosphere coupling irradiance onto a horizontal surface, and it is multiplied by the sky-view factor under the assumption that the atmospheric scattered radiance is isotropic, i.e., diffuse.

In this study, the neighboring slope search radius is set to 1 km, following the typical adjacency range reported in the ATCOR algorithm description [17]. The contribution of surrounding pixels is modeled by convolving their radiance values with a point spread function (PSF), which characterizes the fraction of radiance from neighboring pixels entering the sensor aperture. The PSF is approximated by a Gaussian function, which has been widely used in atmospheric correction algorithms such as FLAASH. This treatment ensures that the influence of neighboring terrain is spatially weighted in a physically consistent manner [40].

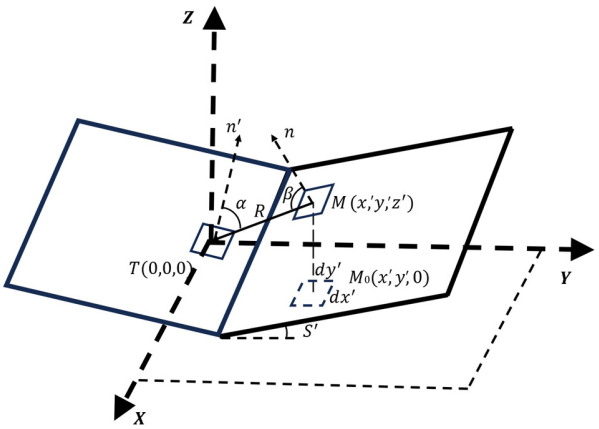

Fig. 2. Calculation of the irradiance from the neighboring slope.

The irradiance from the neighboring slope can be expressed as follows under the single scattering approximation:

$$E^*_{\text{ref}}(x, y, z, \lambda) = V_{\text{terrain}} \cdot \int\int_{(x',y')} \frac{E^*_{\text{dir}}(x', y', z', \lambda) + E^*_{\text{dif}}(x', y', z', \lambda)}{\pi} \cdot \rho(x', y', z', \lambda) \cdot \tau \cdot \frac{\cos\alpha \cdot \cos\beta}{R^2} \cdot \frac{dx'dy'}{\cos S'}. \tag{8}$$

As shown in Fig. 2, establishing a 3-D Cartesian coordinate system centering at the target pixel T and with the XTY plane as horizontal, the two oblique solid-frame planes are the tangent planes at pixels T and M, respectively, R is the distance from the adjacent pixel M to T, τ is the transmittance along the path $\overrightarrow{MT}$, representing the radiative transfer from the neighboring slope pixel M to the target pixel T. α is the acute angle between the line $\overrightarrow{MT}$ and the normal to the surface at T, and β is the acute angle between the line $\overrightarrow{MT}$ and the normal to the surface at M, S' is the slope at M, V_{terrain} is the terrain view factor, which quantifies the fraction of the surrounding terrain that is visible from the perspective of a sloped target pixel T. Its calculation method is given by the following equation [39]:

$$V_{\text{terrain}} = \frac{(1 + \cos S)}{2} - V_{\text{sky}} \tag{9}$$

where S denotes the slope at the pixel.

The path radiance does not contain any ground object information and belongs to noise. The path radiance $L_{\text{atm}}(x, y, z, \lambda)$ is calculated by making look-up tables (LUTs) using MODTRAN 6 [16]. The LUT parameters are the sensor viewing angle and the surface elevation. This is because the surface elevation alters the length of the atmospheric path, which in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

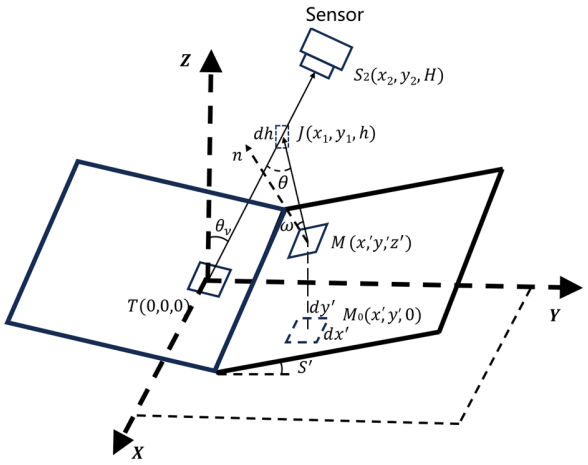

Fig. 3. Calculation of the atmospheric adjacency effect.

turn influences atmospheric backscattering. As a result, it is essential to consider surface elevation when estimating path radiance.

The atmospheric adjacency effect contributes to the spectral mixing and blurs the image. It is modeled under the single scattering approximation as follows, see Fig. 3:

$$L_{a_\mathrm{adj}}(x,y,z,\lambda) = \int\int_{(x',y')} \frac{E^*_{\mathrm{dir}}(x',y',z',\lambda) + E^*_{\mathrm{dif}}(x',y',z',\lambda)}{\pi} \cdot \rho(x',y',z',\lambda) \cdot G(x',y',z',\lambda)\, dx'dy' \tag{10}$$

$$G(x',y',z',\lambda) = \int_N \frac{\zeta(\theta,\lambda)\cdot\cos\theta\cdot\cos\omega}{\left[(x'-x_1)^2+(y'-y_1)^2+(z'-h)^2\right]\cos S'} \cdot \tau_1\left(\overrightarrow{MJ},\lambda\right) \cdot \tau_2\left(\overrightarrow{JS_2},\lambda\right)\cdot\frac{dh}{\cos\theta_v}. \tag{11}$$

Suppose the atmosphere is planar, parallel, and vertically divided into N layers, using the same coordinate system as in Fig. 2. $\rho(x',y',z',\lambda)$ represents the reflectance of the adjacent pixel M. $G(x',y',z',\lambda)$ is the atmospheric influence function. As shown in Fig. 3, J is the scattering point and $(x'-x_1)^2+(y'-y_1)^2+(z'-h)^2$ is the squared value of the distance between the adjacent pixel M and the scattering point J, and the positions of the target and sensor can be determined from the image geometry model (IGM) file which contains geolocation and illumination-observation geometric information on a per-pixel basis, see Section II-C. $\tau_1(\overrightarrow{MJ},\lambda)$ is the transmittance from the surrounding image element to the scattering point, and $\tau_2(\overrightarrow{JS_2},\lambda)$ is the transmittance from the scattering point to the sensor. $\zeta(\theta,\lambda)$ is the angular scattering coefficient. θ is the angle between the normal direction of the target object and the observation line of sight. ω is the angle between the normal direction of the adjacent pixel M and the direction of its reflection. h is the height of the scattering point, and θ_v is the viewing zenith angle.

By transforming (1), we obtain (12), as shown at the bottom of the page, for solving the reflectance from the at-sensor radiance $L(x,y,z,\lambda)$, which $T^{\uparrow}(z,\lambda)$ is the upward transmittance. In this correction model, the radiative components are related to the target pixel's geographical location, terrain, and atmospheric status. The pixel's geographical location and terrain information can be determined through the geolocation process with an IGM file as the result, and the atmospheric status during imaging can be inverted using MODTRAN. Specifically, the transmittance terms τ_1 and τ_1 are derived from MODTRAN 6 outputs under the assumption of a plane-parallel, vertically stratified atmosphere divided into N layers. Specifically, MODTRAN provides the layer-by-layer atmospheric transmittance. For τ_1, the cumulative transmittance is computed from the neighboring surface element M to the scattering point J multiplying the transmittance values of the intervening layers. Similarly, τ_2 is obtained as the product of the transmittance of the atmospheric layers between the scattering point J and the sensor. Therefore, both τ_1 and τ_1 are determined by the layer-wise transmittance along the corresponding optical path. $E^*_{\mathrm{ref}}(x,y,z,\lambda)$, $E^*_{\mathrm{coup}}(x,y,z,\lambda)$, and $L_{a_\mathrm{adj}}(x,y,z,\lambda)$ require not only geographical location and atmospheric conditions but also the surrounding surface reflectance $\rho(x',y',z',\lambda)$. Therefore, it is necessary to calculate the initial value of reflectance under simplified conditions and then solve for the final reflectance through an iterative process.

B. Initial Value of Surface Reflectance

Under the conditions that the surface is Lambertian and the terrain is flat, the initial value of reflectance can be solved. Here, instead of selecting a fixed reflectance value as that in ATCOR [17], we adopt the FLAASH model [13], i.e., (13), to obtain a more accurate initial reflectance value by accounting for the influence of adjacent pixels. This strategy avoids the iteration converging to a local optimal solution and allows for a faster convergence to the final reflectance value, for rugged terrain in subsequent calculations

$$L = \frac{\mathrm{A}\rho_0}{1-\rho_e s} + \frac{\mathrm{B}\rho_e}{1-\rho_e s} + L_a. \tag{13}$$

Here, L represents the at-sensor radiance, ρ_0 represents the target reflectance, ρ_e denotes the average reflectance of adjacent pixels, s represents the atmospheric hemispherical albedo, L_a represents the path radiance, and A and B are coefficients related to the illuminating and atmospheric conditions. The at-sensor radiance is considered as consisting of three components: the portion directly reflected from the target pixel, the portion reflected by the background surface and scattered into the target pixel, and the path radiance. By using a two-step technique, the target reflectance ρ_0 can be derived [41], providing the initial value of surface reflectance.

$$\rho_i = \frac{\pi\left(\left[L(x,y,z,\lambda) - L_{\mathrm{atm}}(x,y,z,\lambda) - L_{a_\mathrm{adj}}(x,y,z,\lambda)\right]\right)}{\left[E^*_{\mathrm{dir}}(x,y,z,\lambda) + E^*_{\mathrm{dif}}(x,y,z,\lambda) + E^*_{\mathrm{coup}}(x,y,z,\lambda) + E^*_{\mathrm{ref}}(x,y,z,\lambda)\right]\cdot T^{\uparrow}(z,\lambda)}. \tag{12}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

The first step is the retrieval of atmospheric status, including cloud, aerosol optical thickness (AOT), and water vapor information. Specifically, cloud screening is performed prior to aerosol and water vapor retrieval. Low- and mid-altitude opaque clouds are identified based on their high reflectance in the visible and near-infrared bands, while thin cirrus clouds are detected using the 1.38-μm water vapor absorption channel, which is highly sensitive to cirrus but largely opaque to surface reflectance. Pixels identified as cloudy are masked out and excluded from subsequent atmospheric parameter retrieval and reflectance inversion. AOT is estimated using the dark-pixel method [42], where surface reflectance at 660 nm (red band) is approximated from MODTRAN 6 simulations in combination with the shortwave infrared (SWIR) band at 2100 nm, which is minimally affected by aerosols. An empirical linear relationship between red and SWIR reflectance is applied (e.g., $\rho_{660\text{nm}} \approx 0.5 \cdot \rho_{2100\text{nm}}$), and a LUT is constructed linking AOT with the mean reflectance of dark pixels across the image. Optimization against this LUT yields the image-wide visibility value.

For column water vapor, pixel-wise retrieval is carried out using MODTRAN-generated LUTs constructed around water vapor absorption features at 820, 940, and 1135 nm. Radiance values from both the absorption bands and adjacent shoulder bands are used to build a 2-D. LUTs, where one axis represents the reflectance-dependent shoulder band value and the other represents the shoulder-to-absorption ratio. Two water vapor estimates are obtained per pixel and averaged to provide the final value.

The second step is the solution of (13) based on these retrieved atmospheric parameters. The low-/mid-altitude opaque cloud mask, cirrus mask, columnar water vapor on a per-pixel basis, and AOT are also generated and output as quality indicators.

C. Geometric Path of Radiative Transfer in Rugged Terrain

Before performing the iterative calculation of the reflectance in rugged terrain, it is necessary to determine the geometric path along which the radiative transfer in the physical model is calculated on a per-pixel basis. This involves the geometric location of the pixel footprint, the terrain parameters, as well as the viewing angles.

The extraction of the above-mentioned geometric parameters primarily relies on digital surface model (DSM) data [43]. It represents the 3-D surface, including natural and man-made objects such as vegetation and buildings.

The pixel-wise determination of footprint latitude, longitude, and elevation is conducted with the DSM data and the hyperspectral radiance image in company with the related rational polynomial coefficients (RPCs) [44], [45]. It is performed by minimizing the discrepancy between observed and modeled pixel image coordinates, through an iterative process, to solve the geographical coordinates of each pixel in the rugged surface [46]. To retain the radiative and geometric information as precisely as possible, the resulting geolocation of each pixel is not directly used to resample the hyperspectral radiance image or the DSM data, but output to a multiband IGM file, in which the first band comprises the longitude of each pixel, the second band the latitude, and the third band the elevation.

Then, the DSM data is used to calculate the terrain factors for each ground pixel, including slope, aspect, the solar incidence angle, shaded factor, sky-view factor, and terrain view factor. And the terrain factors are also stored in the multiband IGM file.

As for the viewing angle $\cos\theta_{v,i}$ related to each ground pixel in rugged terrain, it is similar to the solar incidence angle in the calculation method, specifically using viewing zenith θ_v and azimuth angles ϕ_v instead of solar zenith and azimuth angles in (3). It is also stored in the IGM file as an individual band for foreseeable use in the future, although the surface is taken as Lambertian in this article

$$\cos\theta_{v,i} = \cos S \cos\theta_v + \sin S \sin\theta_v \cos(A - \phi_v). \tag{14}$$

In the above-mentioned way, the per-pixel geometric parameters, including pixel position, illuminating and viewing angles, as well as terrain factors, are inherently co-registered in the coordinate system of the DSM data, which is supposed to be with the highest standard in geolocation accuracy as a mapping product. As a matter of fact, the accuracy of geometric parameters is also affected by that of the RPCs.

D. Iterative Process for Reflectance in Rugged Areas

The solving process of model (12) involves (2)~(11), in which (6)~(8) and (10) rely on the surface reflectance. Therefore, it is an implicit model that need to be iteratively solved. The specific iterative process is as follows.

1) The initial reflectance value ρ_0 is calculated as in Section II-B.
2) Calculate $E^*_{\text{dir}}(x,y,z,\lambda)$, $E^*_{\text{dif}}(x,y,z,\lambda)$, $L_{\text{atm}}(x,y,z,\lambda)$, $T^{\downarrow}(z,\lambda)$, and $T^{\uparrow}(z,\lambda)$ using MODTRAN and (2)~(5). Since MODTRAN is a single-point computational model based on the assumption of a flat terrain, it can be used to construct an LUT of direct solar irradiance and diffuse sky irradiance for different water vapor contents and different elevations. Then, they are interpolated on the water vapor and elevation, combined with the terrain factors, to obtain the corresponding radiative components pixel by pixel. As the AOT has been retrieved in the step of initial reflectance calculation and is assumed as uniform in the whole image based on the dark pixel method, it is not necessarily taken as a variable in the LUT. Furthermore, by combining the terrain factors of each pixel with the flat radiation components, the radiation components of each pixel in rugged terrain can be obtained.
3) Calculate the intermediate values of $E^*_{\text{ref}}(x,y,z,\lambda)$, $E^*_{\text{coup}}(x,y,z,\lambda)$, and $L_{a_\text{adj}}(x,y,z,\lambda)$ using the reflectance ρ_i, and then obtain the reflectance ρ_{i+1} by using (12).
4) Iterate step 3) a few times until the reflectance value converges, and get the final reflectance ρ_t. Extensive experimental testing has shown that convergence can mostly be achieved within three iterations, with fluctuations in reflectance value not exceeding 0.01. This is because the initial value is determined under a flat terrain

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

assumption and on a per-pixel basis, which is already close to the actual reflectance. Therefore, it improves the convergence and its speed. In terms of computational efficiency, the runtime of ICGAT is comparable to iterative correction methods such as ATCOR, while being faster than optimization-based algorithms such as ISOFIT, owing to its rapid convergence and lightweight iterative design.

5) The result hyperspectral reflectance image is in the spatial grids of the radiance image, with the IGM file containing the abundant geometric information necessary for geometric resampling. The resampling is suggested to be conducted as the last step only if the reflectance image needs to be output as a map. In this way, the radiometric information is retained from the interference of geometric sampling as much as possible.

E. Reflectance Uncertainty Quantification Based on GUM

Since it is difficult to conduct field measurement for validation of the reflectance in rugged terrain, especially in areas that are challenging to arrive, e.g., steep slopes, high altitude mountains, top of canopies, and so forth, a method that does not rely on in situ measurement to quantify the reflectance uncertainty is proposed here, as an important part of ICGAT method. It would be valuable for the subsequent application-oriented information extraction from hyperspectral remote sensing data.

The GUM is a series of metrological specifications, which has been generalized for uncertainty assessment of scientific measurement processes [47].

In 2010, the Committee on Earth Observation Satellites (CEOS) developed the Quality Assurance Framework for Earth Observation (QA4EO) [48], which introduced the GUM uncertainty framework into the field of Earth observation to provide a quantitative quality indicator to the data product. The QA4EO emphasizes the fact that physical quantities' values and the associated uncertainties are based on "traceability," involving the relationship of relevant quantities to national or international (community-agreed) standards through an unbroken chain of comparisons. It gives a brief description of the uncertainty evaluation process according to the law of propagation of uncertainty (LPU), resulting in radiance quantities accompanied by uncertainty values.

In addition to the LPU, the Monte–Carlo method (MCM) is also included in the framework of GUM, as a numerical approach in situations where it is difficult or inconvenient to provide the partial derivatives of the model [49]. The MCM conducts the propagation of distributions by performing random sampling from probability distributions.

The uncertainty analysis for the reflectance retrieved by ICGAT uses a combination of MCM and LPU, i.e., by using MCM to determine the uncertainty of an implicit intermediate quantity, which is then used as an input to LPU in the subsequent uncertainty analysis based on an explicit model [50], [51]. Specifically, MCM is employed for intermediate variables that cannot be expressed through an explicit analytical model, such as atmospheric parameter retrievals (e.g., AOT and column water vapor). The resulting uncertainties from MCM subsequently propagate to the estimation of radiance and atmospheric transmittance, and are then further propagated into the framework for surface reflectance uncertainty assessment. For the measurement model (12), LPU [see (15)] generates an estimate of a priori variance for the output quantity ρ_t depending on the input radiance and relevant intermediate quantities

$$u(\rho_t) = \sqrt{\begin{aligned} &\left(\frac{\partial \rho_t}{\partial L}\right)^2 \cdot u(L)^2 + \left(\frac{\partial \rho_t}{\partial L_{\text{atm}}}\right)^2 \cdot u(L_{\text{atm}})^2 \\ &+ \left(\frac{\partial \rho_t}{\partial L_{a_\text{adj}}}\right)^2 \cdot u\left(L_{a_\text{adj}}\right)^2 + \left(\frac{\partial \rho_t}{\partial E^*_{\text{dir}}}\right)^2 \cdot u\left(E^*_{\text{dir}}\right)^2 \\ &+ \left(\frac{\partial \rho_t}{\partial E^*_{\text{dif}}}\right)^2 \cdot u\left(E^*_{\text{dif}}\right)^2 + \left(\frac{\partial \rho_t}{\partial E^*_{\text{coup}}}\right)^2 \cdot u\left(E^*_{\text{coup}}\right)^2 \\ &+ \left(\frac{\partial \rho_t}{\partial E^*_{\text{ref}}}\right)^2 \cdot u\left(E^*_{\text{ref}}\right)^2 + \left(\frac{\partial \rho_t}{\partial T^{\uparrow}}\right)^2 \cdot u\left(T^{\uparrow}\right)^2 \\ &+ 2\sum_{1\le i<j}^{8} \frac{\partial \rho_t}{\partial u_i}\frac{\partial \rho_t}{\partial u_j} r_{ij} u_i u_j \end{aligned}} \tag{15}$$

where $u(\cdot)$ represents the uncertainty of the specific quantity. r_{ij} denotes the correlation coefficient between the ith and jth radiative components, while u_i and u_j represent two of the first eight component uncertainties. In the implementation, the partial derivatives in (15) are computed analytically based on (12). Therefore, the uncertainty propagation process consists of the following steps.

1) *Define Inputs and Outputs:* The model's inputs include at-sensor radiance, DSM, satellite observation metadata (including satellite azimuth, zenith, imaging time, imaging altitude, latitude, longitude, and spatial resolution). The output is surface reflectance.
2) *Identify Uncertain Inputs:* Key sources of uncertainty include the calibration uncertainty of at-sensor radiance, the horizontal and vertical uncertainties of the DSM, and the uncertainty of satellite observation metadata. In addition, uncertainties arise from using radiative transfer models such as MODTRAN.
3) *Determine the Probability Distribution of Uncertain Inputs:* Standard uncertainty is evaluated using a "Type A," i.e., from a probability density function derived from an observed frequency distribution, or "Type B," i.e., from a subjective probability, approach [47]. Specifically, radiometric calibration uncertainty is typically evaluated using a "Type B" approach based on instrument specifications or laboratory calibration reports. DSM horizontal and vertical uncertainties are evaluated using a "Type B" approach based on product documentation. Uncertainties in satellite observation metadata are generally evaluated using a "Type B" approach based on literature and sensor metadata. For parameters derived from radiative transfer models such as MODTRAN, uncertainties are assessed using a "Type B" method based on literature.
4) *Uncertainty Propagation via LPU and MCM:* LPU is applied to processing steps with analytical solutions, while MCM is used for complex iterative calculations. Specifically, geometric uncertainties are handled using MCM, while the uncertainties of terrain

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

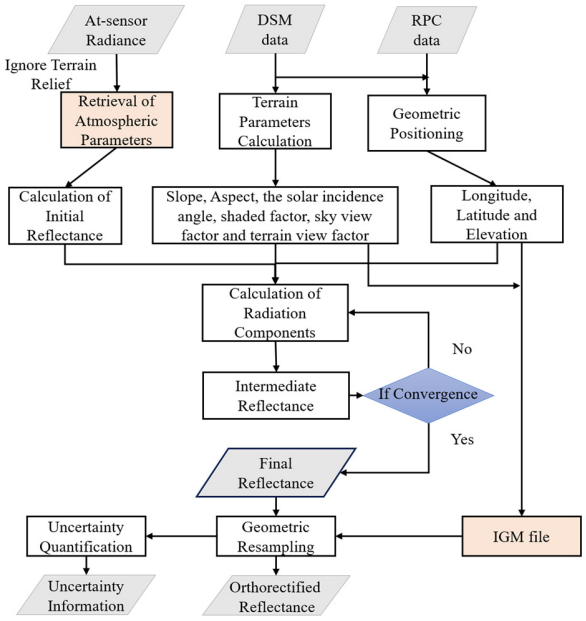

Fig. 4. Flowchart of ICGAT.

parameters—derived from the DSM, such as slope and aspect—are propagated using MCM. For atmospheric parameters (AOT and column water vapor), uncertainties are estimated through MCM simulations that combine the effects of at-sensor radiance uncertainty with the inherent MODTRAN uncertainty. These combined uncertainties, originating from atmospheric, terrain, and geometric factors, are jointly propagated via MCM to all radiative and transmittance components in (15). The resulting radiative component uncertainties are then further propagated through LPU to obtain the overall surface reflectance uncertainty.

5) *Per-Pixel and Per-Band Uncertainty Estimation:* The uncertainty assessment of the final surface reflectance is obtained based on (15) via the LPU.

In summary, the flowchart of ICGAT is shown in Fig. 4. In addition to the orthorectified reflectance, the processing outputs include IGM files, atmospheric parameter files, and uncertainty estimates, enabling quantitative evaluation of reflectance quality and further analysis.

F. Evaluation Approaches

In order to validate the accuracy of the ICGAT algorithm, the simulated data from 3-D radiative transfer models is first used. The advantage of processing with the simulated data is that it allows taking the input reflectance data to the simulation model as the truth for reference, thus enabling a direct validation of the ICGAT result.

Then, the hyperspectral data obtained by three actual payloads, i.e., EnMAP, EMIT, and GF-5 AHSI, are selected to validate the surface reflectance results obtained by the ICGAT method with the official reflectance products of these payloads, respectively.

TABLE I
LESS RADIATIVE TRANSFER SIMULATION PARAMETERS

Parameter	Data
Imaging date	May 22, 2019
Imaging local time	13:15:09
Number of rows	100
Number of columns	100
Number of bands	224
Sensor height	642 km
Pixel size	30 m
Center latitude	40.184796° N
Center longitude	115.718779° E
Atmospheric visibility	42 km

For different data sources, different sites are selected to cover as many as possible the various conditions of ground object classes and degrees of terrain relief.

Meanwhile, multiple surface reflectance retrieval methods are also involved in the comparison. The traditional FLAASH method applicable to flat surfaces and the open-access official PACO module applicable to rugged terrain are involved for EnMAP data, the FLAASH and the Imaging Spectrometer Optimal Fitting (ISOFIT) [52] for EMIT data, and the cascaded approach of terrain correction [53], [54], [55] + FLAASH [56], [57] (hereafter referred to as the cascaded correction method) for GF-5 AHSI data.

The validation is performed through visual inspection, spectral comparison of land cover reflectance on sunny and shaded slopes, comparison of the decorrelation effect of reflectance from terrain effects, as well as comparison of the coefficient of variation (CV) of reflectance.

III. DATA AND RESULTS

A. Simulation Data Obtained From 3-D Radiative Transfer

LESS [58] (large-scale remote sensing data and image simulation framework over heterogeneous 3-D scenes) is used to simulate a virtual scene and the corresponding spectral image. LESS has been widely applied for evaluating topographic correction methods, and its suitability for large-scale heterogeneous 3-D scene simulation has been highlighted in previous studies [22], [59]. Compared with other radiative transfer models such as DART [60], LESS provides an efficient framework for generating satellite-scale benchmark datasets over rugged terrain [58], which makes it particularly suitable for validating the proposed ICGAT framework in this study.

The scene consists of two land cover types: soil and forest. Forest is uniformly and densely distributed in the west (left) half of the image, while the east (right) half is entirely covered by soil, forming a precise 50:50 spatial division, as shown in Fig. 5(a). The elevation range of the input DEM is 311~1048 m, as shown in Fig. 5(b). The average slope is 29.2°, and the maximum slope reaches 64.8°. With the radiative transfer simulation parameters shown in Table I, the radiance image acquired by a virtual hyperspectral imager at 642-km observation altitude is simulated, see Fig. 5(c). The simulation at an altitude of 642 km aims to replicate the imaging

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

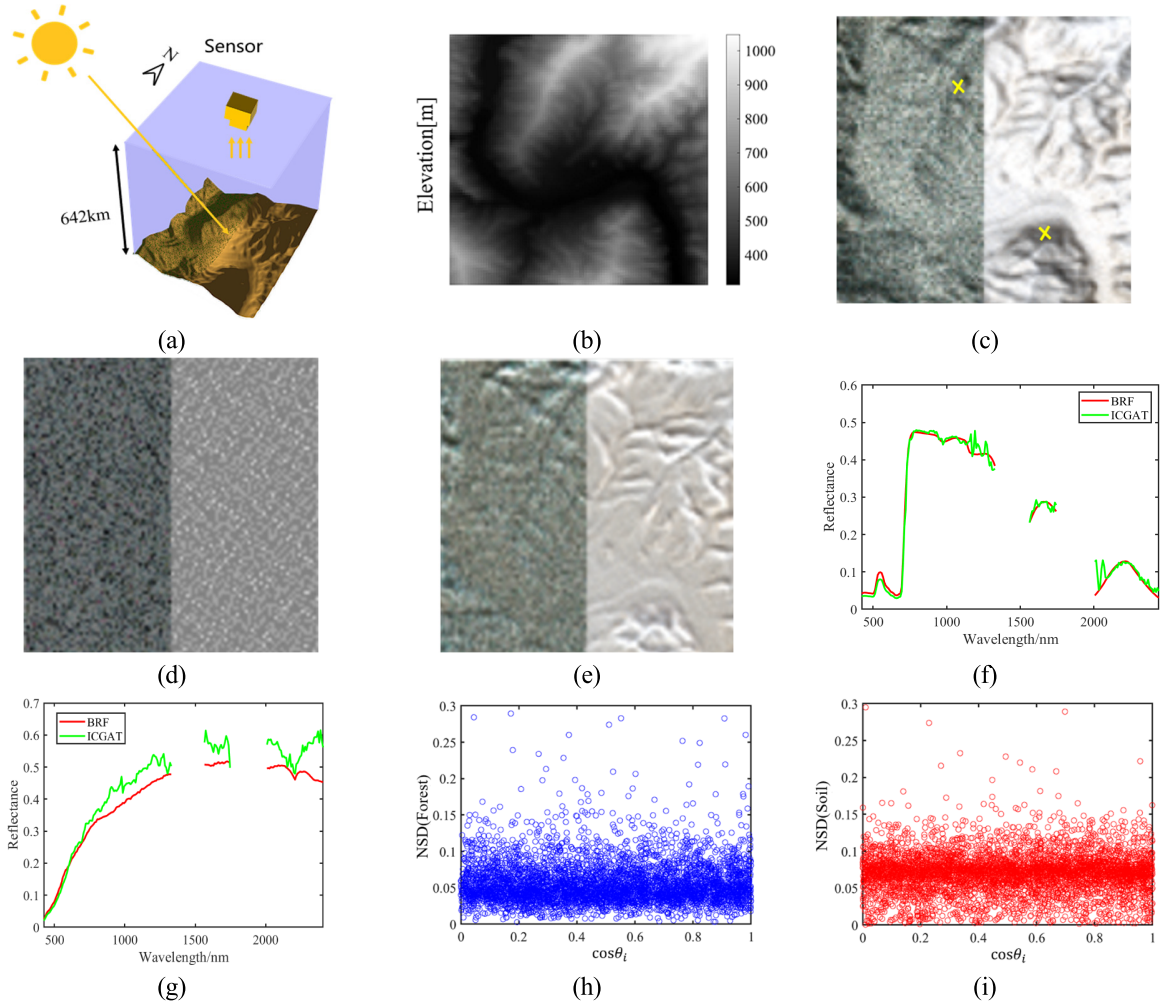

Fig. 5. Validation of ICGAT based on LESS simulation. (a) Schematic of the scene and (b) corresponding DEM. (c)~(e) RGB composite images (R: 641.3 nm, G: 550.7 nm, and B: 459.0 nm) showing radiance image (two shaded pixels of tree and soil land cover type are marked with yellow crosses), BRF image, and ICGAT result image, respectively. (f) and (g) Comparisons of tree and soil reflectance spectra, respectively. Per-pixel NSD scatter plots of (h) forest pixels and (i) soil pixels with respect to the cosine of the solar incidence angle based on the BRF image.

conditions of the EnMAP satellite, with a spatial resolution of 30 m and the same spectral configuration comprising 224 bands. A scene size of 100 × 100 pixels (corresponding to a 3-km swath) is used to ensure both the effectiveness of validation and the inclusion of potential neighboring land cover effects. With reference to Fig. 5(b), it can be seen that in Fig. 5(c) the terrain-induced shadows are clearly visible, with slopes facing away from the sun appearing darker, while those oriented toward the sun appear significantly brighter. By using LESS, the surface reflectance spectral image is generated under the assumption that the surface is flat as a byproduct of radiance image simulation. It is called a BRF image, as shown in Fig. 5(d).

The simulated radiance image under rugged surface conditions is then inverted by the ICGAT algorithm, see Fig. 5(e). It can be seen that the shadow effect is suppressed to some extent. This ICGAT result is finally compared with the BRF image on a flat surface in the sense of spectral consistency, to evaluate the inversion accuracy of ICGAT over terrain relief.

The visual comparison of the reflectance spectra is shown in Fig. 5(f) and (g), for the tree and soil sample pixels marked as crosses in Fig. 5(c), respectively. It can be seen that the ICGAT reflectance spectra match the BRF spectra well overall, except for the slight over-/under-corrections in the water vapor absorption bands around 940 and 1135 nm.

Furtherly, the normalized spectral distance (NSD) is calculated for the tree and soil spectra in Fig. 5(f) and (g), to quantify the spectral similarity between two spectra x_i and y_i [19]

$$\mathrm{NSD} = \sqrt{\frac{\sum_i^N (x_i - y_i)^2}{\sum_i^N \left[\frac{(x_i + y_i)}{2}\right]^2}}. \tag{16}$$

The NSD is 0.059 for the forest sample pixel and 0.136 for the soil sample pixel, which demonstrates that ICGAT reflectance is quite similar to the BRF as a reference.

Furthermore, using the BRF image as a reference, per-pixel NSD scatter plots with respect to the cosine of the solar incidence angle are analyzed separately for forest pixels (left side) and soil pixels (right side), as shown in Fig. 5(h) and (i). The results indicate that both forest and soil pixels

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

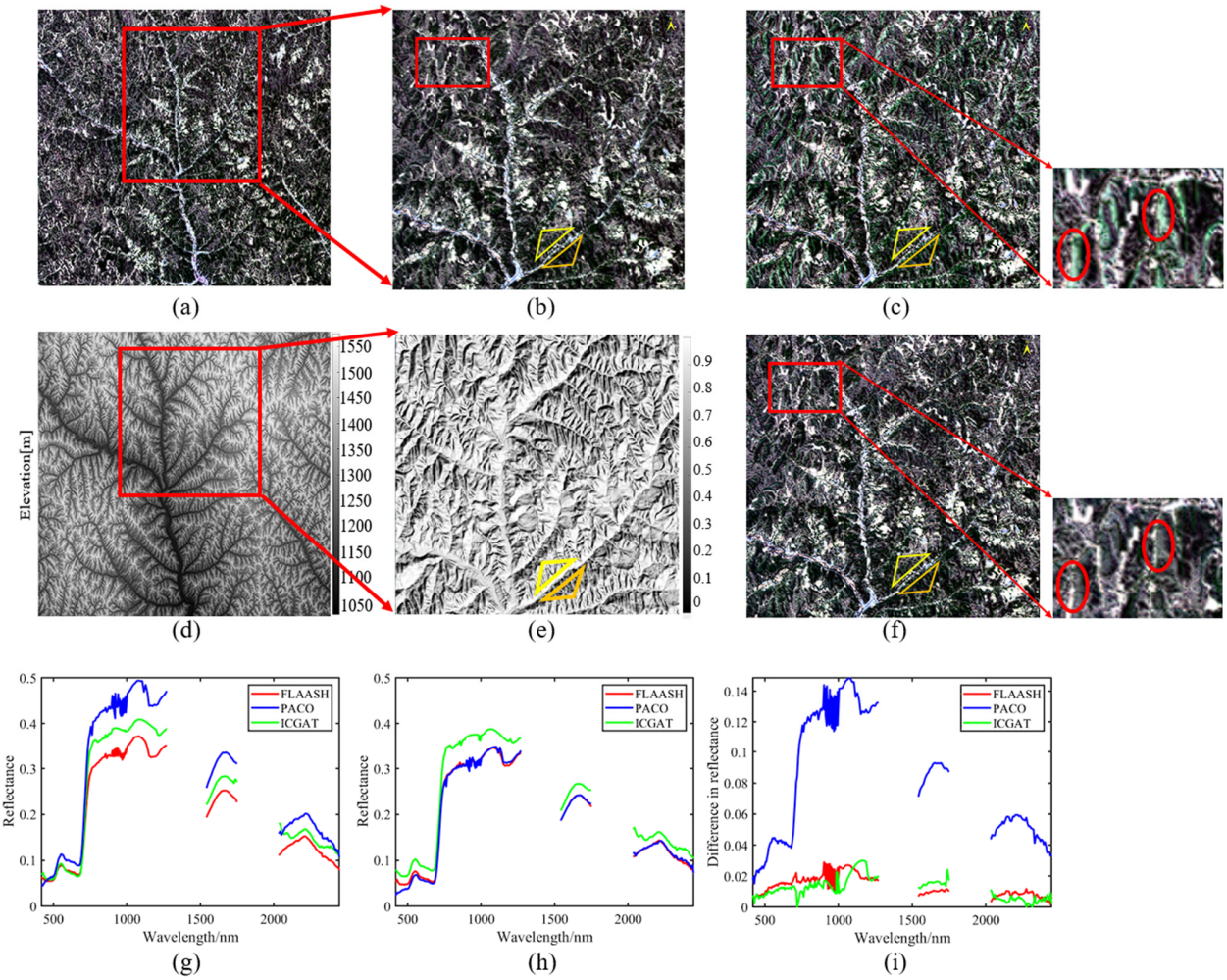

Fig. 6. Validation of ICGAT based on EnMAP data: RGB composite (R: 641.3 nm, G: 550.7 nm, B: 459.0 nm) of (a) radiance image, (b) FLAASH reflectance image for the ROI in (a), and (c) PACO reflectance image for the ROI in (a). (d) Grayscale DSM and (e) cosine of the solar incidence angle for the ROI in (a). (f) RGB composite of ICGAT reflectance image for the ROI in (a). In which (c) and (f) are zoomed-in-view in to compare the over-correction effects. Comparison for the average spectra of the neighboring sunny and shaded slopes: (g) average spectra of the sunny slope marked with the yellow triangle in (b)–(f). (h) Average spectra of the shaded slope marked with the orange triangle. (i) Spectral difference between the neighboring sunny and shaded slopes.

remain relatively stable as the cosine of the solar incidence angle increases, with most NSD values below 0.15. This further confirms the effectiveness of the ICGAT algorithm in suppressing terrain-induced effects.

B. EnMAP Data

The EnMAP L1C data [6] in the Northern Shaanxi Province, China, are downloaded with the specific parameters as shown in Table II. The RGB composite of the hyperspectral radiance image is shown in Fig. 6(a).

The accompanying terrain data is chosen from the ALOS Global DSM "ALOS World 3D—30 m" (AW3D30) product, with an elevation accuracy of 5 m and a horizontal accuracy of 5 m, evaluated in terms of root-mean-square error (RMSE) [61]. It is appropriate to use DSM data to depict the surface that interacts with the light in the radiative transfer process, since it contains height information of surface structures such as trees and artificial infrastructures. The grayscale image of the DSM is shown in Fig. 6(d).

TABLE II
IMAGE PARAMETERS OF ENMAP DATA

Parameter	Data
Imaging date	June 8, 2023
Imaging local time	11:50:20
Number of rows	1569
Number of columns	1236
Number of bands	224
Sensor height	642 km
Pixel size	30 m
Center latitude	36.973° N
Center longitude	109.315° E

The region in the red rectangles in Fig. 6(a) and (d) is selected as the region of interest (ROI) to conduct the validation in the original image scale. It has a typical loess plateau landscape with drastic topographic relief. The elevation range is 1027~1575 m, with the average elevation of 1294 m and a standard deviation of 88 m. The average slope is 22° while the standard deviation of the slope is 9°, with the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

maximum slope up to 57°. The sunny and shaded slopes are fragmented, as shown in Fig. 6(e), making it suitable for testing the performance of the ICGAT algorithm.

Three algorithms, i.e., FLAASH, PACO, and ICGAT, are compared to further illustrate the performance of the ICGAT under conditions of vegetated terrain.

Using the reflectance images obtained by the three different methods, the ROIs of the correction result images are compared as shown in Fig. 6(b), (c), and (f), in which the PACO result is the corresponding official EnMAP L2A product. It can be seen that both PACO and ICGAT effectively suppressed the phenomenon of shadowing, but the PACO result shows the presence of over-corrected light green curves on some vegetated slopes, see the zoomed-in-view area in Fig. 6(c).

A local area composed of neighboring sunny and shaded slopes is selected and marked with yellow and orange triangles, respectively, in Fig. 6(b)–(f), as an example of comparative analysis on terrain-induced reflectance differences. As the two face-to-face slopes are along the same valley stretched in the northeast direction, they are covered by the same type of vegetation. The consistency of the reflectance between the two slopes obtained from the three methods is shown in Fig. 6(g)~(i). It is obvious that the ICGAT algorithm is the most effective one in the sense of decreasing the spectral difference between the neighboring sunny and shaded slopes of the same vegetation cover, followed by FLAASH and finally PACO. In addition, the NSD between the spectra of the same type of vegetation, on neighboring sunny and shaded slopes in Fig. 6(g) and (h), reflects this situation as well. The NSD obtained by the ICGAT method is 0.051, by FLAASH 0.069, and by PACO 0.35.

The relatively poor performance of PACO in this experiment is primarily due to over-correction on sunny slopes, as shown by the highest reflectance value in Fig. 6(g). It is probably caused by a mismatch between the DEM and EnMAP data. Because the Copernicus DEM data used in EnMAP L2A production is acquired between 2011 and 2015, there is a strong likelihood of temporal inconsistency. This particularly affects vegetated areas as a result of the vegetation growth period. Notably, these limitations are scenario-specific, and the EnMAP L2A product generally shows high terrain correction accuracy.

To quantitatively evaluate the efficacy of the correction methods, the linear correlation between the cosine of the solar incidence angle image and each of the reflectance images is calculated, respectively. Because it is an important indicator for the effect of direct solar irradiance change induced by slope [62], the Pearson correlation coefficient [63] is calculated and plotted as in Fig. 7(a). It can be seen that both the ICGAT and PACO algorithms reduced the correlation coefficient to below 0.05 for all the bands, effectively overcoming the influence of the slope effect on the direct solar illumination.

To further illustrate the decrease of terrain-induced effects on the reflectance, the CV of all the vegetation pixels' reflectance is calculated. The standard deviation σ divided by the average μ, is used to quantitatively describe the spectral variability [64]

$$\mathrm{CV} = \sigma/\mu. \tag{17}$$

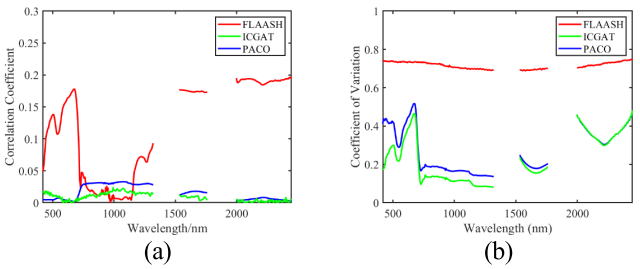

Fig. 7. Statistical analysis of reflectance correction results as a function of wavelength for EnMAP. (a) Correlation coefficients between the reflectance image and the cosine of the solar incidence angle. (b) CV of reflectance for all the vegetation pixels.

TABLE III
IMAGE PARAMETERS OF EMIT DATA

Parameter	Data
Imaging date	Oct 15, 2024
Imaging local time	14:37:00
Number of rows	1280
Number of columns	1242
Number of bands	285
Sensor height	400 km
Pixel size	60 m
Center latitude	37.59645° N
Center longitude	119.6835° W

The result can be observed in Fig. 7(b). The CV of the ICGAT result in VNIR is slightly smaller than that of the PACO result, while the CV of the FLAASH result is obviously higher.

C. EMIT Data

The experimental site selected for EMIT data is located on the western side of the Sierra Nevada in CA, USA. The Sierra Mountains run through it, and the terrain has significant undulations. The parameters of EMIT data are listed in Table III.

A local region of EMIT images is selected for comparative analysis due to its distinct topographic variability, shown as the rectangle in Fig. 8(a). This area exhibits significant elevation changes and varied illumination conditions, making it an ideal candidate for evaluating the correction methods on reflectance consistency.

The corresponding AW3D30 DSMs are shown in Fig. 8(b). The elevation range in this region is 229~1302 m, with an average elevation of 674 m and a standard deviation of 211 m. The average slope is 13°, with the maximum up to 49° and the standard deviation 8°. The cosine of the solar incidence angle image is shown in Fig. 8(c).

Three algorithms, i.e., FLAASH, ISOFIT, and ICGAT, are compared for the reflectance of uncovered terrain, as shown in Fig. 8(d)~(f). The visual effect of the ICGAT reflectance image is flatter than that of the other two methods, as it compensates for the sunny and shadow effects in the mountainous regions. As can be seen from Fig. 8(g) to (i), ICGAT effectively suppresses the difference in reflectance between the neighboring sunny and shaded slopes, which is much lower

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

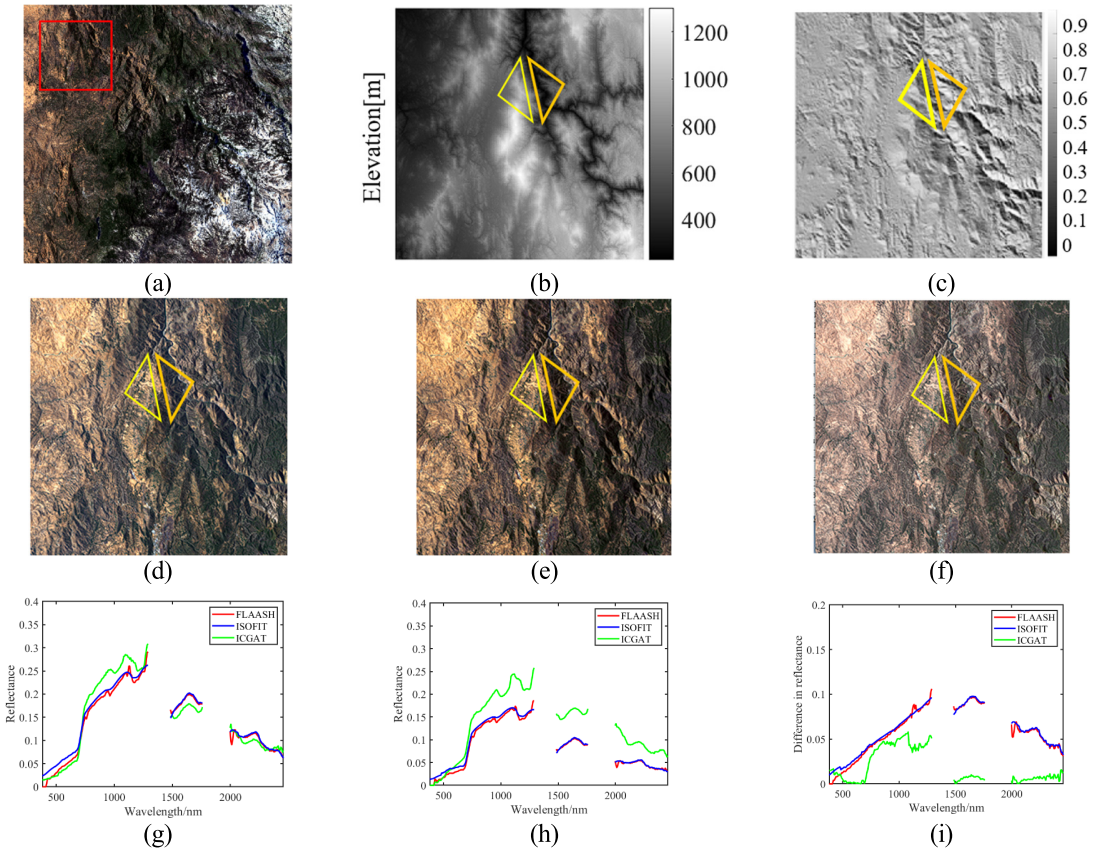

Fig. 8. True-color EMIT image, DSM data, and reflectance correction results (a) RGB composite (R: 641.3 nm, G: 551.9 nm, and B: 462.6 nm) radiance image, in which the red rectangle shows the local region corresponding to the following subfigures, (b) grayscale DSM, and (c) cosine of the solar incidence angle including the ROI. (d)~(f) Reflectance results from three correction methods: FLAASH, ISOFIT, and ICGAT, respectively. Comparison for the average spectra of the neighboring sunny and shaded slopes: (g) average spectra of the sunny slope marked with the yellow triangle in (b)~(f), (h) average spectra of the shaded slope marked with the orange triangle, and (i) spectral difference between the neighboring sunny and shaded slopes.

than that obtained by the other two methods. The NSD results are as follows: the result of ICGAT is 0.1699, which is lower than both 0.4889 of ISOFIT and 0.4947 of FLAASH. This reveals that the spectral similarity between the neighboring sunny and shaded slopes is mostly promoted by ICGAT.

The Pearson correlation coefficient between the cosine of the solar incidence angle and the reflectance is computed, and the result is presented in Fig. 9(a). It can be seen that, compared with the ISOFIT and FLAASH algorithms, the ICGAT algorithm effectively reduces the correlation coefficient. As a result, it achieves the decorrelation between the reflectance and the terrain influence.

Similar to Section III-B, the CV of reflectance is calculated and shown in Fig. 9(b). For wavelengths lower than 655 nm, the CV of ISOFIT is the smallest. From 655 nm on, the reflectance stability of the ICGAT method is the best. The reason is that ISOFIT obtains the AOT parameter for each pixel through an optimal estimation method, which has a more significant impact on a shorter wavelength.

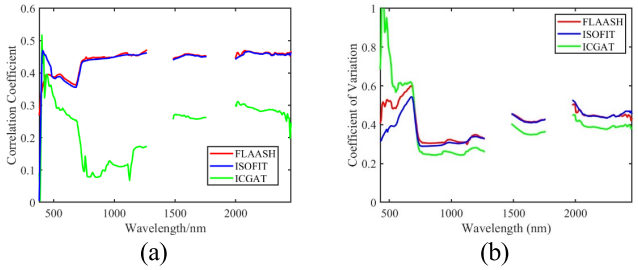

Fig. 9. Statistical analysis of reflectance correction results as a function of wavelength for EMIT. (a) Correlation coefficients between the reflectance and the cosine of the solar incidence angle. (b) CV of reflectance for the whole image.

D. GF-5 AHSI Data

The experimental site selected for GF-5 AHSI data is the Guanting Reservoir in Hebei Province, China, which is characterized by the presence of water bodies, forest, cropland, and cloud land cover types. The parameters of Level 1 AHSI data used in this correction are listed in Table IV.

The corresponding AW3D30 DSM is shown in Fig. 10(a). Statistical analysis reveals that elevation ranges from 35 to 2419 m in this region, with an average elevation of 742 m and a standard deviation of 374 m. The average slope is 17°, and the standard deviation is 13°, while the maximum slope reaches 72°.

Fig. 10(b) shows the RGB composite of the spectral radiance image. There are reservoirs and sandy lands in the north,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

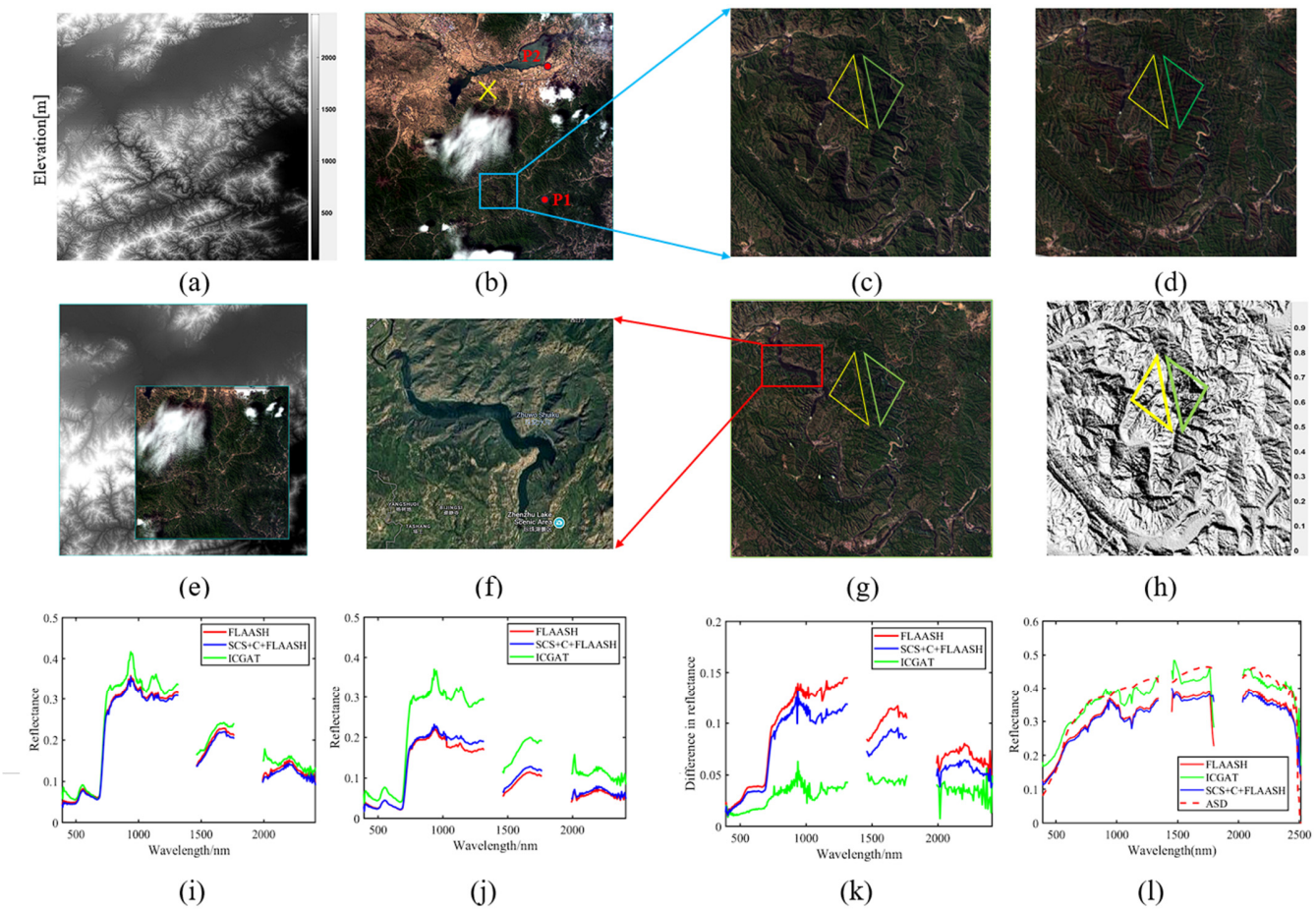

Fig. 10. True-color composite of GF-5 hyperspectral image and grayscale image of DSM data. (a) Grayscale image of DSM data. (b) True-color composite radiance image (R: 638.6 nm; G: 548.7 nm; and B: 458.8 nm), with P1 and P2 indicating the vegetation and sand pixels subsequently validated, respectively. (e) Scroll maps of at-sensor radiance and DSM after geometric positioning. Reflectance of the three methods (c) FLAASH, (d) SCS + C + FLAASH, (g) ICGAT, and (h) Cosine of solar incidence angle. (f) Zhenzhu Lake Scenic Area Google Earth image. Comparison for the average reflectance of neighboring sunny and shaded slopes (i) spectra of sunny slope marked with the yellow triangle in (g), (j) spectra of shaded slope marked with the green triangle in (g), and (k) spectral difference between the neighboring sunny and shaded slopes. (l) Validation of the three methods based on the in situ ASD spectrum for the sand pixel marked with a yellow cross in (b).

TABLE IV
IMAGE PARAMETERS OF GF-5 AHSI DATA

Parameter	Data
Imaging date	May 22, 2019
Imaging local time	13:15:09
Number of rows	2008
Number of columns	2083
Number of bands	327
Sensor height	705 km
Pixel size	30 m
Center latitude	40.184796° N
Center longitude	115.718779° E

with relatively flat terrain. In the south, there are forests with obvious rugged terrain and distinct sunny and shaded slopes.

Fig. 10(e) shows the registration of the radiance data to the DSM data after the geometric positioning step in ICGAT. The subpixel registration accuracy provides a prerequisite for realizing high-accuracy correction of terrain-induced effects subsequently.

The results of the ICGAT algorithm are compared with those of the FLAASH and SCS + C + FLAASH cascaded method.

For the forest area shown as the blue rectangle region in Fig. 10(b), the results of ICGAT and the other two methods are zoomed-in-view in as shown in Fig. 10(c), (d), and (g). The corresponding cosine of the solar incidence angle is depicted in Fig. 10(h). They clearly demonstrate the scene flattening effect achieved by the ICGAT method, eliminating the dependence of image brightness and spectra on slope. On the other hand, the FLAASH results still exhibit pronounced topographic relief. The cascaded method also generates under-corrections in shaded areas, although it has made some compensation for the terrain effects. This advantage of ICGAT is especially evident in the Zhenzhu Lake Scenic Area, which becomes more distinguishable due to accurate correction of the surrounding terrain influences, as shown in Fig. 10(g) and validated by the related Google Earth image Fig. 10(f).

The differences in reflectance spectra obtained by the three methods on neighboring sunny and shaded slopes of the same vegetation are compared, as shown in Fig. 10(i)~(k). It can be seen that the difference spectrum related to ICGAT gets rid

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

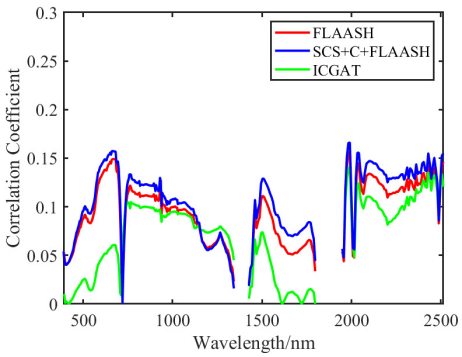

Fig. 11. Correlation coefficients between the reflectance images and the cosine of the solar incidence angle as a function of wavelength.

of the spectral characteristics of vegetation, which indicates a dramatic improvement in the consistency of spectral features between neighboring sunny and shaded slopes.

The NSD between neighboring sunny and shaded slopes for the forest validation zone is calculated as well. Specifically, the NSD obtained by the FLAASH method is 0.5702, the cascaded method 0.4622, and the ICGAT method 0.1538. It shows that ICGAT reduces the under-correction of shaded slopes, significantly improving spectral similarity between neighboring sunny and shaded slopes of the same vegetation landcover.

Except for the comparison between the shaded and sunny slopes, the reflectance results are furtherly validated with reference to the field measurement data obtained from an analytical spectra devices (ASDs) spectro-radiometer in a relatively flat sand site near Guanting Reservoir, as shown by the yellow cross in Fig. 10(b). As shown in Fig. 10(l), the reflectance amplitude obtained by the ICGAT method is the closest to the field spectrum. The NSD between the reflectance results of the three methods and the in situ spectrum is as follows: 0.2437 for FLAASH, 0.2499 for SCS + C + FLAASH, and 0.1982 for ICGAT. Therefore, the ICGAT reflectance is the closest to the ground reference.

The Pearson correlation coefficients between each band of the reflectance image and the cosine of the solar incidence angle are shown in Fig. 11. The correlation coefficients related to ICGAT decreased relative to those related to the other two methods. So the IGGAT method has the best effect in decorrelation of reflectance from topography.

To evaluate the impact on remote sensing applications, an unsupervised k-means classification is conducted on the reflectance images, setting four classes and a maximum of 100 iterations. These four classes are interpreted as water bodies, forests, cropland, and clouds, based on comparison with Google Earth images. From Fig. 12, it can be seen that the ICGAT-derived result successfully suppresses the confusion between cloud and cropland, and achieves the best continuous spatial distribution of forest class, getting the best consistency with the visual perception. Therefore, it is better than those of the other two methods. This classification experiment also serves as a representative downstream validation, showing that the reflectance products retrieved by ICGAT provide more reliable inputs for land cover mapping compared with other correction methods.

Furthermore, CV is calculated for all the forest pixels and cropland pixels, respectively, as shown in Fig. 13. It can be seen that the overall CV for forest and cropland cover types decreased in the ICGAT result, indicating reduced intraclass spectral variability.

E. Evaluation of the Uncertainty of the Hyperspectral Surface Reflectance

This section shows how to quantify the uncertainty of the reflectance by using GF-5 AHSI data over Guanting Reservoir as an example. Since the ICGAT model, i.e., (12), has a nonlinear iterative solving process, the LPU and MCM are combined through the following steps.

1) Define the Inputs and Outputs of the Model: The inputs are GF-5 AHSI radiance and corresponding DSM data, as well as satellite observation metadata. The outputs are surface reflectance.

2) Decide Which Inputs Are Treated as Uncertain: In this study, key sources of uncertainty include the radiometric calibration accuracy of at-sensor radiance, the horizontal and vertical accuracy of the DSM, the accuracy of satellite observation metadata, and the use of MODTRAN for atmospheric radiative transfer. For atmospheric parameters, the uncertainties of column water vapor and AOT are derived from their respective retrieval processes, based on Monte–Carlo simulations [49]. These uncertainties are then propagated as inputs for subsequent calculations of radiance and atmospheric transmittance uncertainty. The uncertainty traceability chain diagram (Fig. 14) has been redrawn strictly in accordance with the Fidelity and Uncertainty in Climate data records from Earth Observations (FIDUCEOs) framework [65] to explicitly illustrate the full propagation process.

3) Derive the Probability Distribution of the Uncertain Inputs: Since the reflectance obtained during a single imaging process is the target output, a "Type B" evaluation of standard uncertainty is used, which is based on the manufacturer's specification documented in the system requirements. According to the GF-5 AHSI radiometric calibration literature, the uncertainty of radiance can be considered to be 5% (assuming a Gaussian distribution) [1].

According to the accuracy analysis literature of AW3D30 DSM, the vertical and horizontal accuracy are both around 5 m [61], which also varies with different terrain relief conditions. According to the GUM, it is assumed to follow a Gaussian distribution with a coverage factor $k_G = 1.96$ at the 95% coverage probability, so the vertical uncertainty of the DSM is estimated to be $u(p) = 5$ m/1.96. Similarly, the horizontal uncertainty of the DSM can be estimated as $u(q) = 5$ m/1.96.

Regarding the uncertainty of satellite observation metadata, as reported in [66], the geometric positioning accuracy of the GF-5 AHSI data is within 40 m. Accordingly, it is assumed to follow a Gaussian distribution with a coverage factor $k_G = 1.96$ at the 95% coverage probability, so the geometric positioning uncertainty for GF-5 AHSI is estimated to be 40 m/1.96. The uncertainties associated with other geometric parameters are considered negligible, as they are less likely to significantly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

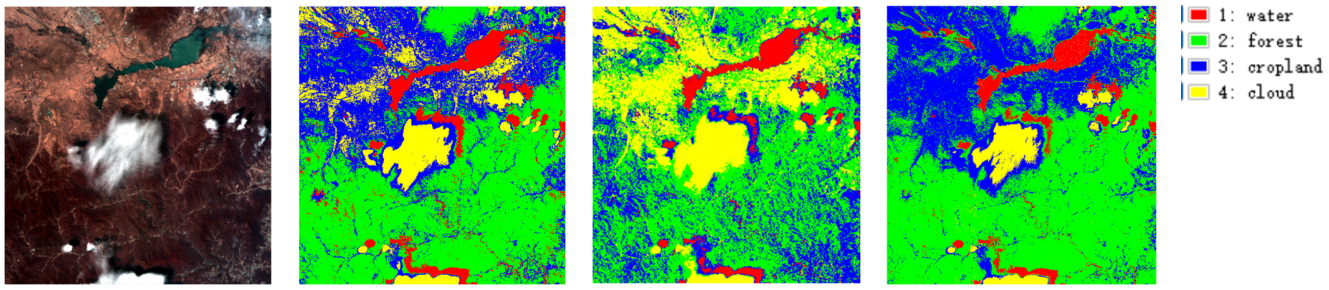

Fig. 12. Comparison of unsupervised k-means classification of different reflectance retrieval methods. (a) Radiance image. Classification of (b) FLAASH, (c) SCS + C + FLAASH, and (d) ICGAT results.

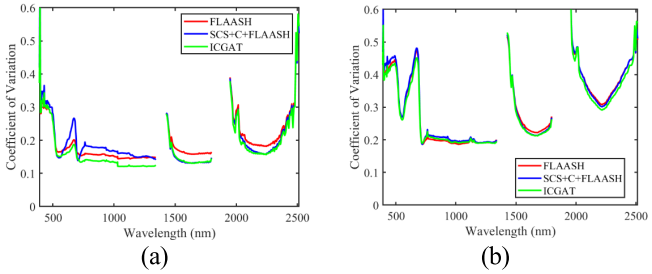

Fig. 13. CV comparison (a) forest and (b) cropland.

impact the overall accuracy given the specific focus of this study.

In Fig. 14, the uncertainty of MODTRAN is decomposed into three components: u(AOT), u(WV), and u(0). The first two represent the uncertainties of AOT and column water vapor, respectively, while u(0) denotes the intrinsic model uncertainty of MODTRAN. According to the MODTRAN user manual [16], the uncertainty of simulated radiance is approximately 1%, and that of transmittance is 0.005, assuming a Gaussian distribution. For the atmospheric parameters (AOT and water vapor), uncertainties are quantified through Monte–Carlo simulations of their respective retrieval processes. Since both parameters are derived from at-sensor radiance, their uncertainties are estimated by combining the uncertainty of the radiance with the algorithmic uncertainties of the retrieval scheme. Specifically, 3000 Monte–Carlo simulations are performed to obtain robust estimates of their variability, which is sufficient to ensure that the change in the estimated standard deviation between successive simulations falls below 1%, indicating statistical convergence. For completeness, we have further illustrated the uncertainty traceability and retrieval process for AOT and WV in the Appendix.

In summary, the sources and values of uncertainty considered in this study are listed in Table V.

Based on the Monte–Carlo simulations, the uncertainty of AOT is estimated as a scene-level constant value of 0.009 for the GF-5 AHSI data over Guanting Reservoir, reflecting the spatially averaged variability across all dark-pixel-based retrievals. In contrast, the column water vapor is retrieved on a per-pixel basis, and its uncertainty is therefore computed pixel-wise. The average uncertainty of column water vapor over the entire scene is 0.066 g/cm^2.

TABLE V
SOURCES AND QUANTIFICATION OF UNCERTAINTY

Uncertainty Source	Uncertainty Value	Probability Distribution Function
At-sensor Radiance	5%	Gaussian
DSM Vertical Uncertainty	$5m$/1.96	Gaussian
DSM Horizontal Uncertainty	$5m$ / 1.96	Gaussian
Positioning Accuracy	$40m$/1.96	Gaussian
The intrinsic model uncertainty of MODTRAN	Simulated Radiance: 1% Transmittance: 0.005	Gaussian

4) Uncertainty Propagation to the Surface Reflectance on a Per-Pixel and Per-Band Basis by Using LPU and MCM: Based on Table V, the uncertainties of the atmospheric parameters (AOT and water vapor content), together with the intrinsic MODTRAN uncertainty, the positioning accuracy, and the uncertainties of terrain parameters (slope, aspect, cosine of the solar incidence angle, shaded factor, and sky-view factor derived from DSM), are jointly propagated through Monte–Carlo simulations. Specifically, 3000 Monte–Carlo iterations are conducted to simulate all radiative components in (15), with each iteration sampling from the corresponding uncertainty distributions of these input variables. This number of iterations is sufficient to ensure that the change in the estimated standard deviation between successive simulations falls below 1%, indicating statistical convergence. The standard deviation of the simulated results is then used to represent the uncertainty of each radiative term, while the correlations among radiative components are evaluated using their pairwise covariance matrices. The partial derivatives in (15) are analytically computed based on (12). Finally, the overall surface reflectance uncertainty is synthesized using the LPU according to (15), providing the per-pixel and per-band uncertainty estimates.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

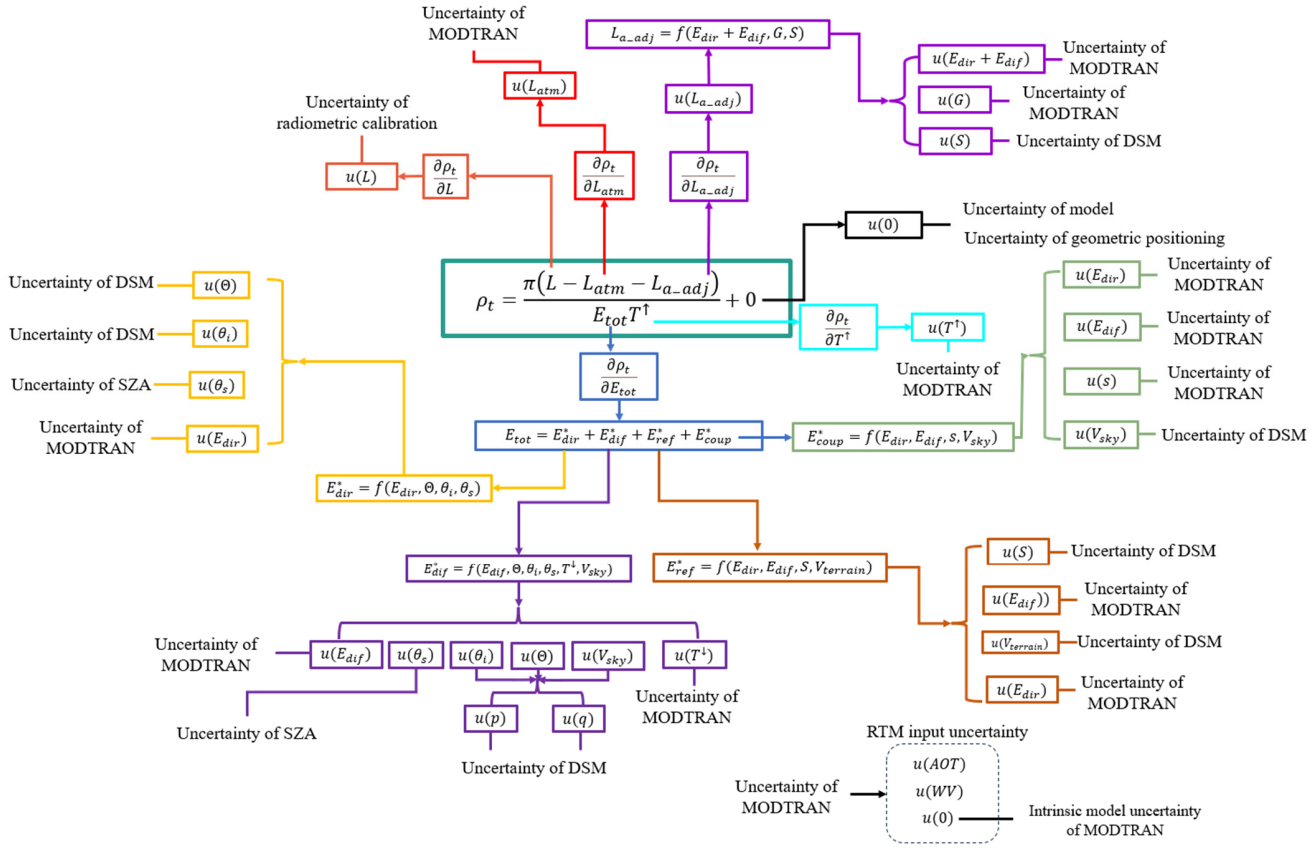

Fig. 14. Diagram of the uncertainty propagation chain.

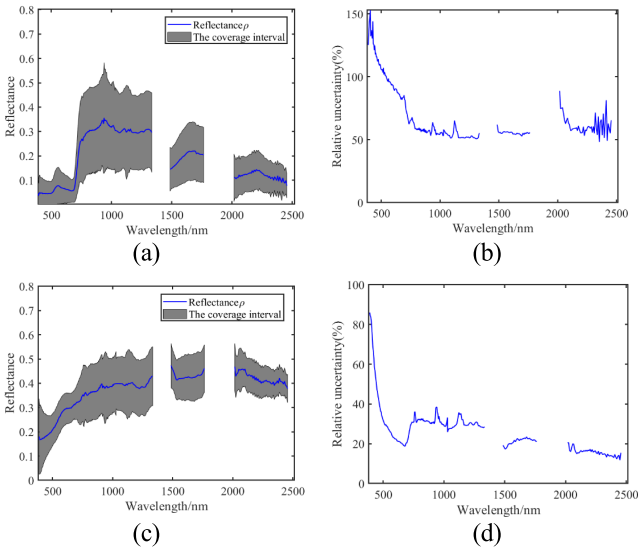

Fig. 15. Reflectance uncertainty obtained using LPU combined with MCM. (a) Vegetation reflectance and 99.7% probability coverage interval. (b) Standard relative uncertainty of the vegetation reflectance. (c) Soil reflectance and 99.7% probability coverage interval. (d) Standard relative uncertainty of the soil reflectance.

5) Validation of the Surface Reflectance Uncertainty: For validation, the reflectance and its uncertainty for one vegetation pixel [as labeled P1 in Fig. 10(b)] are shown in Fig. 15(a) and (b), while the reflectance and its uncertainty for one soil pixel [P2 in Fig. 10(b)] are shown in Fig. 15(c) and (d) (Gaussian distribution, coverage factor $k = 3$). Fig. 15(b) and (d) represents the standard relative uncertainty, calculated as the ratio of the standard uncertainty to the estimated reflectance $u(\rho_t)/\rho_t$.

Generally speaking, the standard relative uncertainty decreases gradually with increasing wavelength and eventually stabilizes. The relative uncertainty of vegetation is relatively large, stabilizing at around 52% after 500 nm, while the uncertainty of soil is smaller, stabilizing at under 35%. Reflectance uncertainty of soil is generally low due to the flat terrain of that site. In contrast, vegetated areas in rugged terrain exhibit higher uncertainty, primarily caused by terrain-induced variations in local solar illumination angles and shadowing.

In addition, the difference between the in situ measured and the inversed sand reflectance, related to the spectra in Fig. 10(l), is shown with the uncertainty in Fig. 16. The light gray and dark gray zones in the figure represent the coverage interval related to the standard uncertainty and expanded uncertainty, i.e., with 68% and 99.7% coverage probability, respectively. It can be seen that the reflectance differences are within the coverage interval that contains the sand reflectance truth with a 99.7% probability; thus, the reflectance and uncertainty results are cross-validated in the relatively flat sand site.

IV. Discussions

By all the above results, the ICGAT method has demonstrated its effectiveness in multiple aspects.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

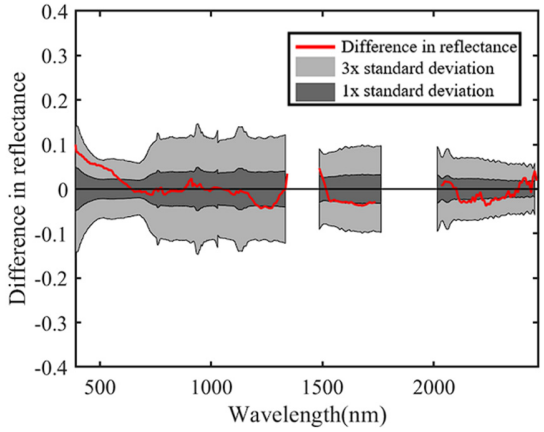

Fig. 16. Comparison of retrieved and in situ sand reflectance difference and corresponding uncertainty magnitude (light gray: three standard deviations and dark gray: one standard deviation).

1) Comparisons between the ICGAT and state-of-the-art atmospheric and terrain correction methods demonstrate that the ICGAT method exhibits superior performance in reducing terrain effects, particularly in enhancing the spectral consistency of the same land cover on neighboring sunny and shaded slopes, i.e., reducing the intraclass spectral variability of reflectance.
2) In terms of the correlation between reflectance image and topography, the ICGAT method exhibits the lowest overall correlation across most spectral bands.
3) Regarding landcover classification, the ICGAT method improves classification accuracy.
4) Last but not least, by combining the LPU and MCM methods along the ICGAT algorithm processing chain, it gives an uncertainty budget for surface reflectance, which provides a tool to quantify the reflectance quality in rugged terrain.

Other than in situ measurement in a relatively flat area, it is very difficult and lacks of widely accepted protocol to obtain high-quality field reflectance as ground truth in rugged areas. Therefore, in the following, uncertainty analysis techniques are furtherly discussed about the way to determine the main uncertainty sources of inverse reflectance of rugged surfaces, and the extent to which DEM/DSM data quantitatively affect the inversion quality.

A. Analysis of the Main Uncertainty Sources of Reflectance

In terms of quantifying the quality of remote sensing reflectance data, Origo et al. [67] suggested that atmospheric correction uncertainty assessment products should be further developed and released with surface reflectance products, and passed on to derivative products to inform users of the uncertainty. They analyzed the uncertainty of multispectral reflectance data by using MCM. Thompson et al. [68], [69] carried out airborne and ground simultaneous experiments to validate inverted AVIRIS reflectance and corresponding uncertainty. Storch et al. [70] validated EnMAP reflectance

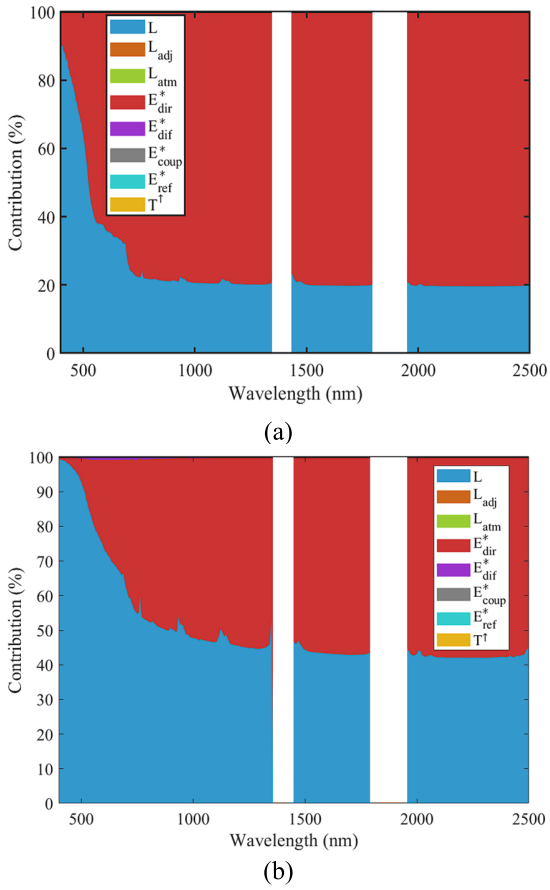

Fig. 17. Uncertainty contribution from each radiative component (a) vegetation pixel and (b) soil pixel.

products by comparing the posterior uncertainty, which is the RMSE of the products with reference to in situ measured data, with the uncertainty budget of satellite design. However, these uncertainty quantification methods are unable to accomplish the uncertainty propagation. They are not applicable to sensitivity analysis to identify the main uncertainty source and to enhance the data quality subsequently.

In the coupled MCM and LPU framework, an uncertainty budget (such as an added column in Table V) that quantifies individual source contributions to the overall surface reflectance uncertainty is infeasible. Nevertheless, the contributions of individual radiative components in (15) to the total reflectance uncertainty can still be analyzed. In order to ascertain the main factors affecting the reflectance products' quality, the uncertainty contribution from the components in (15) is compared, as shown in Table VI. The contribution factors for the vegetation and soil pixels, as in Fig. 15, are drawn in Fig. 17. It stacks the ratios of the uncertainty component to the total uncertainty, which sum to one.

Fig. 17 reveals the following information for the specific pixels' reflectance.

1) The direct solar irradiance and the at-sensor radiance are the two major influencing factors, accounting for 99% of the reflectance uncertainty, except for the strong

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
UNCERTAINTY CONTRIBUTION OF EACH COMPONENT

At-sensor radiance	Path radiance	Atmospheric adjacency effect	Direct solar irradiance
$\frac{(\frac{\partial \rho_t}{\partial L})^2 \cdot u(L)^2}{u^2(\rho_t)}$	$\frac{(\frac{\partial \rho_t}{\partial L_{atm}})^2 \cdot u(L_{atm})^2}{u^2(\rho_t)}$	$\frac{(\frac{\partial \rho_t}{\partial L_{a_{adj}}})^2 \cdot u(L_{a_{adj}})^2}{u^2(\rho_t)}$	$\frac{(\frac{\partial \rho_t}{\partial E^*_{dir}})^2 \cdot u(E^*_{dir})^2}{u^2(\rho_t)}$
Diffuse sky radiance	**Surface-atmosphere coupling irradiance**	**Irradiance from neighboring slope**	**Atmospheric transmittance**
$\frac{(\frac{\partial \rho_t}{\partial E^*_{dif}})^2 \cdot u(E^*_{dif})^2}{u^2(\rho_t)}$	$\frac{(\frac{\partial \rho_t}{\partial E^*_{coup}})^2 \cdot u(E^*_{coup})^2}{u^2(\rho_t)}$	$\frac{(\frac{\partial \rho_t}{\partial E^*_{ref}})^2 \cdot u(E^*_{ref})^2}{u^2(\rho_t)}$	$\frac{(\frac{\partial \rho_t}{\partial T})^2 \cdot u(T)^2}{u^2(\rho_t)}$

absorption bands of water vapor (around 1400 and 1900 nm).

2) The uncertainty contribution ratio of the at-sensor radiance is dominant in the purple and blue spectral range, smaller for longer wavelengths, and stable after 700 nm, which strongly correlated with the wavelength-dependent scattering characteristics of the aerosols. This indicates the importance of accurate treatment of aerosol in both radiometric calibration and surface reflectance inversion processes.
3) After about 550 nm, the uncertainty contribution ratio of the direct solar irradiance is dominant, which indicates that the uncertainty of the terrain data and its co-registration with the radiance image should be focused on to further suppress the reflectance uncertainty.
4) Compared with the vegetation pixel, the contribution of radiance uncertainty is more pronounced in the soil pixel. This is reasonably due to the higher radiance magnitude and flatter terrain, which amplifies the impact of radiometric accuracy while reducing the influence of solar illumination geometry.

Therefore, the uncertainty of this reflectance dataset mainly traces back to the hyperspectral radiance image and the DSM data uncertainty. Their weights vary with wavelength and the degree of terrain relief. It should be noted that the uncertainty propagation and major sources analysis need to be conducted specifically for each dataset, considering the specific value of uncertainty sources. In addition, pixels with extremely high uncertainty (e.g., >50%) may need to be flagged or down-weighted in subsequent applications such as vegetation index retrieval or land cover classification, in order to prevent error amplification. Moreover, the dominant uncertainty sources are expected to vary across scenarios—for example, radiometric calibration errors may be more influential to flat, bright surfaces, whereas DSM-related errors may play a larger role in steep terrains. A systematic decomposition of these contributions through sensitivity analysis and scenario-specific evaluations will be an important focus of our future work, which will enhance the interpretability and usability of the uncertainty information for downstream applications.

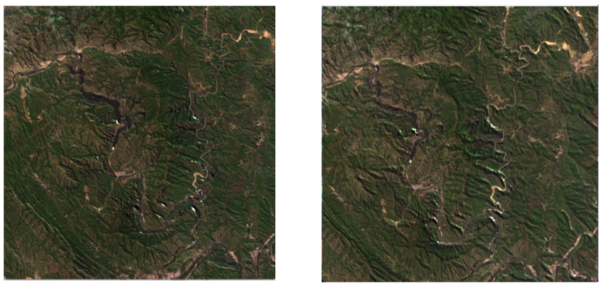

Fig. 18. Reflectance obtained using different DSM products in the localized region (a) Copernicus DEM and (b) ASTER GDEM.

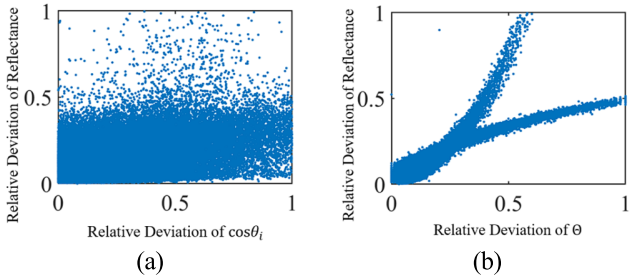

Fig. 19. Scatter plot of correlation between relative deviation of reflectance and relative deviation of terrain parameters (a) $\cos(\theta_i)$ and (b) shaded factorΘ.

B. Effect of DSM Data Accuracy on the Inverse Reflectance

Compared with the radiometrically recalibration of the hyperspectral radiance image, improvement to the choice of DSM data is easier to implement. Therefore, it is further discussed as follows.

In Section III-D, AW3D30 DSM data have been used. As control examples, the Copernicus DEM [71] and ASTER GDEM V3 [72], actually both as DSM data, are used instead, respectively. The result reflectance images are shown in Fig. 18(a) and (b). With regard to Fig. 10(g), the result from using the Copernicus DEM is very similar, while there are more and much significant over-corrections and under-corrections in the result from using the ASTER GDEM. This may be attributed to the vertical and horizontal accuracy of the DSM data, which are ~5-m vertical/~5-m horizontal as of AW3D30, ~4-m vertical in terms of LE90/~6-m horizontal in terms of CE90 as of Copernicus DEM [73], and ~10-m vertical/~5-m horizontal as of ASTER GDEM [72], [74], respectively. The accuracy of the ASTER GDEM is lower than that of the AW3D30 DSM and Copernicus DEM.

In addition, taking the correction by using AW3D30 as the reference, the correction by using ASTER is assessed. Scatter plots of relative deviation of reflectance and relative deviation of cosine of solar incidence angle, as well as scatter plots of relative deviation of reflectance and relative deviation of shaded factor are drawn in Fig. 19. It can be seen that the relative deviation of the cosine of the solar incidence angle has a linear correlation with the relative deviation of the reflectance, while the relative deviation of the shaded factor has a strong linear correlation with the relative deviation of reflectance. This reveals that the accuracy of the DSM data

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

mainly affects the calculation of the solar incidence angle and, particularly, the shaded factor, which in turn affects the magnitude of the direct solar irradiance component that is one of the major influencing factors for the retrieved reflectance. This provides a mechanistic explanation for the phenomenon that there are more and much significant over-corrections and under-corrections in the result from using the ASTER GDEM, i.e., Fig. 18(b).

In summary, accurate reflectance inversion in rugged terrain requires accurate characterization of the surface geometry, which should properly account for the land covers. Carmon et al. [35] have conducted preliminary work in estimating surface shape from the radiance measurement and a priori information about atmospheric and geometric parameters, under the optimal estimation framework. This kind of atmospheric–terrain correction method, which does not wholly depend on the accuracy of the input DSM data, is also one of the future development directions.

C. Limitation of the Lambertian Assumption and Future Extensions

In the present study, the ICGAT framework adopts the Lambertian surface assumption, which simplifies the retrieval process by neglecting anisotropic reflectance effects. This assumption is acceptable for surfaces with relatively weak angular dependence, such as bare soils, short vegetation, or impervious urban materials. However, for highly anisotropic targets such as dense vegetation canopies, snow, or complex structured surfaces, neglecting BRDF may introduce additional uncertainties in reflectance estimation. Several studies have demonstrated that BRDF and terrain effects are inherently coupled, as slope and aspect directly influence the incident and exitant angles of radiation. For example, Li et al. [75] developed a physics-based atmospheric and BRDF correction method that incorporates MODIS BRDF products to account for directional surface reflectance, showing improved accuracy in rugged areas. In addition, Hao et al. [25] proposed an improved topography-coupled kernel-driven (TCKD) reflectance model, which corrects diffuse skylight effects by integrating the diffused-equivalent slope model (dESM) with the RossThick–LiTransit (RTLT) kernel-driven framework. They also provided a comprehensive review of BRDF correction methodologies, with particular emphasis on characterizing anisotropic surface reflectance over rugged terrain [76]. These studies indicate that the Lambertian simplification, while computationally efficient, poses a limitation of the current ICGAT framework in scenarios dominated by anisotropic surfaces.

Building upon these findings, future research will focus on extending the ICGAT framework to explicitly incorporate BRDF parameterizations within the core radiative transfer equations. A promising direction is to introduce semi-empirical models, such as the Ross–Li kernel-driven model, to represent directional reflectance characteristics in a computationally efficient manner [77]. By combining such BRDF formulations with the terrain-modified radiative transfer scheme, the model could capture the bidirectional reflectance effects induced by varying solar and viewing geometries in rugged terrain. This integration is expected to further improve correction accuracy over vegetated and snow-covered landscapes, where strong anisotropy is prevalent.

In addition, another current limitation lies in the retrieval of AOT. Neglecting terrain and adjacency effects may introduce biases in mountainous or heterogeneous regions. Previous studies have demonstrated the potential of multiangle-based approaches to improve aerosol retrieval under complex terrain [78], [79]. However, such algorithms require multiangular observation data, which are generally not applicable to the hyperspectral datasets discussed in this study. Moreover, the dark-target-based retrieval currently used provides a scene-wide uniform AOT value. Since the ICGAT framework is developed under the Lambertian surface assumption, the impact of AOT uncertainty on surface reflectance accuracy is relatively minor compared to the direct reflectance variations introduced by terrain-induced illumination differences. In addition, coupling BRDF effects and aerosol retrieval over rugged terrain would substantially increase computational complexity and reduce processing efficiency. Therefore, terrain-dependent AOT retrieval has not been implemented at this stage. Future extensions of ICGAT could incorporate terrain- and adjacency-correction modules into the aerosol retrieval process, enabling more accurate estimation of AOT in rugged landscapes and further enhancing the robustness of the framework.

Collectively, extending ICGAT with BRDF-topography coupling, terrain-aware aerosol retrieval, and uncertainty assessment will establish a more comprehensive correction framework, broadening its applicability to diverse land cover and illumination conditions.

Beyond these physical refinements, future work will also focus on validating ICGAT under challenging atmospheric conditions such as heavy haze and high humidity. These scenarios are particularly relevant for applications in urban, industrial, or tropical regions, where strong aerosol loading and water vapor variability may significantly affect radiative transfer. Incorporating such cases into the evaluation framework will help ensure that ICGAT remains robust and reliable across both complex terrains and diverse atmospheric environments, similar to how Thompson et al. [80] extended their optimal estimation framework to challenging atmospheres.

Finally, improving the computational efficiency of ICGAT represents another important avenue for development. Future work will explore adaptive iteration schemes, in which convergence criteria are dynamically adjusted according to local terrain conditions (e.g., slope and aspect), so that pixels in rugged terrain receive stricter convergence checks while flatter surfaces can converge more quickly. Parallelization of the radiative transfer calculations will also be pursued to further improve runtime performance. These strategies are expected to enhance computational efficiency while preserving correction accuracy, making ICGAT more practical for large-scale and near-real-time applications.

V. Conclusion

This study proposes an ICGAT, i.e., ICGAT, on hyperspectral remote sensing images. By establishing a coupled radiative transfer model between the rugged surface and the atmosphere, it effectively corrects the atmosphere–terrain-induced influences on direct solar illumination/shadow, diffuse sky

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

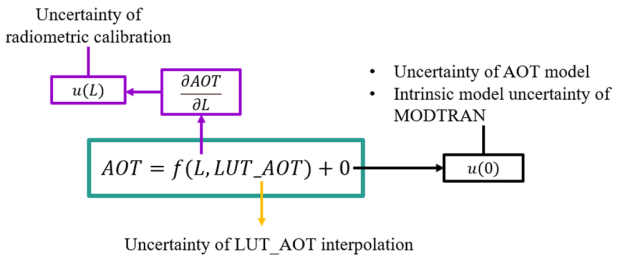

Fig. 20. Uncertainty propagation diagram for AOT retrieval.

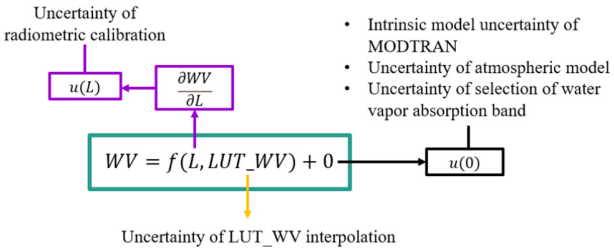

Fig. 21. Uncertainty propagation diagram for column water vapor retrieval.

irradiance, and surrounding background reflection, and reconstructs the hyperspectral surface reflectance characteristics. It then generates regular grid pixels based on the geometric parameters about illumination, geolocation, and observation, which are derived in the correction process and stored in an IGM file. Especially, it also conducts a rigorous uncertainty propagation to the surface reflectance based on GUM, and thus provides a way to identify the main uncertainty sources of the reflectance.

Different testing data and performance parameters are used to evaluate the effectiveness of the proposed ICGAT method. Compared with the state-of-the-art PACO, ISOFIT, SCS + C + FLAASH, and FLAASH methods, the ICGAT method conforms better to the actual radiative transfer over rugged surfaces and effectively improves the accuracy of the inverted reflectance. It obviously overcomes the problem of over- or under-correction caused by terrain relief and enhances the reflectance consistency of the same land cover on neighboring slopes. The unique uncertainty propagation mechanism for the reflectance is a useful tool to identify the main uncertainty source for validation of, and subsequent improvement to, the reflectance in rugged terrain, where field measurement is very difficult to conduct. Meanwhile, the ICGAT method also achieved good results on flat surfaces, which has been validated on a sand site. It could beneficially support subsequent land cover classification and identification.

APPENDIX

This appendix supplements Sections II-D and III-E by detailing the uncertainty traceability for AOT and WV retrieval. Since both parameters are derived from at-sensor radiance using MODTRAN-based LUTs, their uncertainties originate from radiometric calibration, algorithmic retrieval uncertainties, and intrinsic MODTRAN simulation uncertainty. To obtain statistically stable estimates, 3000 Monte–Carlo simulations are performed for each parameter, ensuring convergence by requiring that the change in the estimated standard deviation between successive trials remains below 1%. The corresponding traceability structures for AOT and WV are summarized in Figs. 20 and 21, respectively.

Fig. 20 illustrates the simplified uncertainty propagation diagram for AOT retrieval. The uncertainty in AOT primarily originates from four sources: 1) uncertainty of radiometric calibration, since AOT is retrieved from dark-target radiance and therefore inherits radiometric calibration errors; 2) uncertainty of AOT model, which stems from the assumptions and approximations inherent in the dark-object-based retrieval scheme, as well as from the choice of aerosol type specified in MODTRAN; 3) LUT interpolation uncertainty, arising from numerical interpolation within the MODTRAN-generated AOT LUTs, corresponding to the "LUT_AOT" term shown in Fig. 20; and 4) intrinsic MODTRAN model uncertainty, consistent with the description in Section III-E. These factors collectively contribute to the overall variability of the retrieved AOT.

To obtain a practical and consistent estimate of AOT uncertainty, only the dominant input—uncertainty of radiometric calibration—is explicitly propagated in this study. In each Monte–Carlo iteration, the at-sensor radiance is perturbed according to its uncertainty distribution, AOT is rederived from the LUTs, and the standard deviation of the 3000 realizations is taken as the AOT uncertainty. The number of simulations is sufficient to ensure that the change in the estimated standard deviation between successive iterations falls below 1%, indicating statistical convergence. This Monte–Carlo approach implicitly captures the combined effects of radiance uncertainty, retrieval variability, and part of the LUT and model uncertainties.

Fig. 21 illustrates the simplified uncertainty propagation diagram for WV retrieval. The uncertainty in WV primarily originates from four sources: 1) uncertainty of radiometric calibration, because the retrieval relies on radiance measurements from both strong absorption bands and adjacent shoulder bands; 2) uncertainty of the selection of water vapor absorption bands (820, 940, and 1135 nm), as different band combinations exhibit varying sensitivity to atmospheric moisture and spectral noise; 3) uncertainty of atmospheric model, since different standard atmospheric profiles (e.g., mid-latitude summer/winter, tropical, and subarctic summer/winter) contain distinct vertical temperature–humidity structures, leading to variations in simulated absorption features and thus introducing additional uncertainty into the retrieval; 4) LUT interpolation uncertainty, introduced by numerical interpolation within the MODTRAN-generated WV LUTs, corresponding to the "LUT_WV" term shown in Fig. 21; and 5) intrinsic MODTRAN model uncertainty, consistent with the description provided in Section III-E. These factors jointly contribute to the variability in the pixel-wise water vapor retrievals.

To obtain a practical and statistically stable estimate of WV uncertainty, this study explicitly propagates the dominant input—radiometric calibration uncertainty—through a Monte–Carlo scheme. In each iteration, the radiance values at absorption and shoulder bands are perturbed according to their uncertainty distributions, WV is rederived from the corresponding LUTs, and the standard deviation of 3000

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

realizations is taken as the WV uncertainty. The number of simulations is sufficient to ensure that the change in the estimated standard deviation between successive iterations falls below 1%, indicating statistical convergence. It is noted that WV retrieval can, in principle, be performed using multiple absorption features (e.g., 820, 940, and 1135 nm). In this study, we use only the 940-nm absorption band for computational simplicity. Although this band-selection strategy may introduce additional uncertainty, we consider the simplification reasonable because the 940-nm feature is the most widely used and provides the highest water-vapor sensitivity among the available absorption bands. This Monte–Carlo procedure implicitly captures the combined effects of radiance variability, retrieval algorithm sensitivity, and part of the LUT and model uncertainties.

In summary, the uncertainty analyses for both AOT and WV retrieval follow a consistent metrological framework in which the dominant and quantifiable input—radiometric calibration uncertainty—is propagated through Monte–Carlo simulations to obtain statistically stable uncertainty estimates. Although additional sources such as aerosol-model assumptions, LUT interpolation, MODTRAN inherent errors, and water-vapor band-selection effects may further contribute to the total uncertainty, these factors are either implicitly captured during the Monte–Carlo perturbations or reasonably simplified to maintain tractability. The resulting AOT and WV uncertainties provide consistent, convergence-validated inputs for the subsequent uncertainty propagation in the surface reflectance retrieval, ensuring full traceability from atmospheric parameter estimation to the final reflectance uncertainty budget.

ACKNOWLEDGMENT

The "ALOS World 3D—30 m" digital surface model (DSM) is available freely from https://www.eorc.jaxa.jp/ALOS/en/aw3d30/data/index.htm

The EnMAP L1C and L2A data products made available by the German Aerospace Center (DLR) are accessed through the EnMAP Data Access Portal (https://planning.enmap.org/ips/app#userportal) and processed in compliance with the mission's Acceptable Use Policy.

The EMIT data made available by the NASA Jet Propulsion Laboratory (JPL) are accessed through the EMIT data portal (https://earth.jpl.nasa.gov/emit/data/data-portal/coverage-and-forecasts/).

REFERENCES

[1] Y.-N. Liu et al., "The advanced hyperspectral imager: Aboard China's GaoFen-5 satellite," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 4, pp. 23–32, Dec. 2019, doi: 10.1109/MGRS.2019.2927687.

[2] H. Tang, C. Xiao, K. Shang, T. Wu, and Q. Li, "Radiometric calibration of GF5–02 advanced hyperspectral imager based on RadCalNet Baotou site," *Remote Sens.*, vol. 15, no. 9, p. 2233, Apr. 2023, doi: 10.3390/rs15092233.

[3] L. Yinnian et al., "Development of visible and short-wave infrared hyperspectral imager onboard GF-5 satellite," *Nat. Remote Sens. Bull.*, vol. 24, no. 4, pp. 333–344, Jan. 2020, doi: 10.11834/jrs.20209196.

[4] H. Tang, J. Xie, X. Dou, H. Zhang, and W. Chen, "On-orbit vicarious radiometric calibration and validation of ZY1–02E thermal infrared sensor," *Remote Sens.*, vol. 15, no. 4, p. 994, Feb. 2023, doi: 10.3390/rs15040994.

[5] R. O. Green et al., "The Earth surface mineral dust source investigation: An Earth science imaging spectroscopy mission," in *Proc. IEEE Aerosp. Conf.*, Big Sky, MT, USA, Mar. 2020, pp. 1–15, doi: 10.1109/AERO47225.2020.9172731.

[6] L. Guanter et al., "The EnMAP spaceborne imaging spectroscopy mission for Earth observation," *Remote Sens.*, vol. 7, no. 7, pp. 8830–8857, Jul. 2015, doi: 10.3390/rs70708830.

[7] K. Alonso et al., "Data products, quality and validation of the DLR Earth sensing imaging spectrometer (DESIS)," *Sensors*, vol. 19, no. 20, p. 4471, Oct. 2019, doi: 10.3390/s19204471.

[8] S. Cogliati et al., "The PRISMA imaging spectroscopy mission: Overview and first performance analysis," *Remote Sens. Environ.*, vol. 262, Sep. 2021, Art. no. 112499, doi: 10.1016/j.rse.2021.112499.

[9] J. Nieke and M. Rast, "Towards the Copernicus hyperspectral imaging mission for the environment (CHIME)," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Valencia, Spain, Jul. 2018, pp. 157–159, doi: 10.1109/IGARSS.2018.8518384.

[10] C. M. Lee et al., "An introduction to the NASA hyperspectral InfraRed imager (HyspIRI) mission and preparatory activities," *Remote Sens. Environ.*, vol. 167, pp. 6–19, Sep. 2015, doi: 10.1016/j.rse.2015.06.012.

[11] B.-C. Gao, M. J. Montes, C. O. Davis, and A. F. H. Goetz, "Atmospheric correction algorithms for hyperspectral remote sensing data of land and ocean," *Remote Sens. Environ.*, vol. 113, pp. S17–S24, Sep. 2009, doi: 10.1016/j.rse.2007.12.015.

[12] B.-C. Gao, K. B. Heidebrecht, and A. F. H. Goetz, "Derivation of scaled surface reflectances from AVIRIS data," *Remote Sens. Environ.*, vol. 44, nos. 2–3, pp. 165–178, May 1993, doi: 10.1016/0034-4257(93)90014-o.

[13] G. P. Anderson et al., "MODTRAN4-based atmospheric correction algorithm: FLAASH (fast line-of-sight atmospheric anal-ysis of spectral hypercubes)," *Proc. SPIE*, vol. 4725, pp. 65–71, Aug. 2002, Art. no. 12, doi: 10.1117/12.478737.

[14] F. Kruse, "Comparison of ATREM, ACORN, and FLAASH atmospheric corrections using low-altitude AVIRIS data of Boulder, CO," Jet Propulsion Lab., Pasadena, CA, USA, Tech. Rep., Mar. 2004. [Online]. Available: https://popo.jpl.nasa.gov/pub/docs/workshops/04_docs/Kruse-JPL2004_ATM_Compare.pdfmymargin{

[15] Z. Qu, B. C. Kindel, and A. F. H. Goetz, "The high accuracy atmospheric correction for hyperspectral data (hatch) model," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1223–1231, Jun. 2003, doi: 10.1109/TGRS.2003.813125.

[16] A. Berk, P. Conforti, R. Kennett, T. Perkins, F. Hawes, and J. van den Bosch, "MODTRAN 6: A major upgrade of the MODTRAN radiative transfer code," in *Proc. 6th Workshop Hyperspectral Image Signal Processing: Evol. Remote Sens. (WHISPERS)*, Jun. 2014, pp. 1–4, doi: 10.1109/WHISPERS.2014.8077573.

[17] R. Richter and D. Schläpfer, "Atmospheric and topographic correction (ATCOR theoretical background document)," DLR/ReSe, Cologne, Germany, Tech. Rep. DLR-IB 564-03/2019, 2019. [Online]. Available: https://www.rese-apps.com/pdf/atcor_atbd.pdf

[18] R. de los Reyes et al., "PACO: Python-based atmospheric correction," *Sensors*, vol. 20, no. 5, p. 1428, Mar. 2020, doi: 10.3390/s20051428.

[19] F. Santini and A. Palombo, "Physically based approach for combined atmospheric and topographic corrections," *Remote Sens.*, vol. 11, no. 10, p. 1218, May 2019, doi: 10.3390/rs11101218.

[20] S. Kobayashi and K. Sanga-Ngoie, "A comparative study of radiometric correction methods for optical remote sensing imagery: The IRCvs.Other image-based C-correction methods," *Int. J. Remote Sens.*, vol. 30, no. 2, pp. 285–314, Nov. 2008, doi: 10.1080/01431160802356781.

[21] S. Kobayashi and K. Sanga-Ngoie, "The integrated radiometric correction of optical remote sensing imageries," *Int. J. Remote Sens.*, vol. 29, no. 20, pp. 5957–5985, Sep. 2008, doi: 10.1080/01431160701881889.

[22] Y. Ma et al., "A universal physically-based topographic correction framework for high-resolution optical satellite data," *ISPRS J. Photogramm. Remote Sens.*, vol. 227, pp. 459–480, Sep. 2025, doi: 10.1016/j.isprsjprs.2025.05.027.

[23] W. Li et al., "A model based on spectral invariant theory for correcting topographic effects on vegetation canopy reflectance," *Remote Sens. Environ.*, vol. 322, May 2025, Art. no. 114695, doi: 10.1016/j.rse.2025.114695.

[24] W. Yu, H. Huang, Q. Liu, and J. Wang, "Integrating physical model and image simulations to correct topographic effects on surface reflectance," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 356–371, May 2024, doi: 10.1016/j.isprsjprs.2024.04.017.

[25] D. Hao, J. Wen, Q. Xiao, D. You, and Y. Tang, "An improved topography-coupled kernel-driven model for land surface anisotropic reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2833–2847, Apr. 2020, doi: 10.1109/TGRS.2019.2956705.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

[26] R. Chen et al., "Topographic correction of optical remote sensing images in mountainous areas: A systematic review," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 4, pp. 125–145, Dec. 2023, doi: 10.1109/MGRS.2023.3311100.

[27] R. Richter and D. Schläpfer, "Geo-atmospheric processing of airborne imaging spectrometry data. Part 2: Atmospheric/topographic correction," *Int. J. Remote Sens.*, vol. 23, no. 13, pp. 2631–2649, Jan. 2002, doi: 10.1080/01431160110115834.

[28] Z. Zhang, G. He, X. Zhang, T. Long, G. Wang, and M. Wang, "A coupled atmospheric and topographic correction algorithm for remotely sensed satellite imagery over mountainous terrain," *GISci. Remote Sens.*, vol. 55, no. 3, pp. 400–416, Sep. 2017, doi: 10.1080/15481603.2017.1382066.

[29] D. S. S. Baboo and M. R. Devi, "Geometric correction in recent high resolution satellite imagery: A case study in Coimbatore, Tamil Nadu," *Int. J. Comput. Appl.*, vol. 14, no. 1, pp. 32–37, Jan. 2011, doi: 10.5120/1808-2324.

[30] D. Schläpfer and R. Richter, "Geo-atmospheric processing of airborne imaging spectrometry data. Part 1: Parametric orthorectification," *Int. J. Remote Sens.*, vol. 23, no. 13, pp. 2609–2630, Jan. 2002, doi: 10.1080/01431160110115825.

[31] D. Schläpfer, S. Bojinski, M. Schaepman, and R. Richter, "Combination of geometric and atmospheric correction for AVIRIS data in rugged terrain," Jet Propulsion Lab., Pasadena, CA, USA, Tech. Rep. JPL Publ. 00.18, Feb. 2000. [Online]. Available: http://aviris.jpl.nasa.gov/proceedings/workshops/00_docs/Schlapfer_web.pdf

[32] Z. Ma, G. Jia, M. E. Schaepman, and H. Zhao, "Uncertainty analysis for topographic correction of hyperspectral remote sensing images," *Remote Sens.*, vol. 12, no. 4, p. 705, Feb. 2020, doi: 10.3390/rs12040705.

[33] D. R. Thompson, V. Natraj, R. O. Green, M. C. Helmlinger, B.-C. Gao, and M. L. Eastwood, "Optimal estimation for imaging spectrometer atmospheric correction," *Remote Sens. Environ.*, vol. 216, pp. 355–373, Oct. 2018, doi: 10.1016/j.rse.2018.07.003.

[34] N. Carmon et al., "Unified topographic and atmospheric correction for remote imaging spectroscopy," *Frontiers Remote Sens.*, vol. 3, Jul. 2022, Art. no. 916155, doi: 10.3389/frsen.2022.916155.

[35] N. Carmon et al., "Shape from spectra," *Remote Sens. Environ.*, vol. 288, Apr. 2023, Art. no. 113497, doi: 10.1016/j.rse.2023.113497.

[36] Y. Iikura, "Topographic effects observed in shadowed pixels in satellite imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Toulouse, France, Jun. 2003, pp. 3489–3491, doi: 10.1109/IGARSS.2002.1027225.

[37] H. Zhao, C. Jiang, G. Jia, and D. Tao, "Simulation of hyperspectral radiance images with quantification of adjacency effects over rugged scenes," *Meas. Sci. Technol.*, vol. 24, no. 12, Nov. 2013, Art. no. 125405, doi: 10.1088/0957-0233/24/12/125405.

[38] T. Theoharis, G. Papaioannou, and E.-A. Karabassi, "The magic of the Z-buffer: A survey," in *Proc. Int. Conf. Central Eur. Comput. Graph. Visualizat.*, vol. 23, Jan. 2001, pp. 379–386. [Online]. Available: https://dspace5.zcu.cz/bitstream/11025/11293/1/Theoharis.pdf

[39] J. Dozier and J. Frew, "Rapid calculation of terrain parameters for radiation modeling from digital elevation data," *IEEE Trans. Geosci. Remote Sens.*, vol. 28, no. 5, pp. 963–969, Aug. 1990, doi: 10.1109/36.58986.

[40] G. W. Felde, G. P. Anderson, J. A. Gardner, S. M. Adler-Golden, M. W. Matthew, and A. Berk, "Water vapor retrieval using the FLAASH atmospheric correction algorithm," *Proc. SPIE*, vol. 5425, p. 357, Aug. 2004, doi: 10.1117/12.542513.

[41] G. Jia, A. Hueni, D. Tao, R. Geng, M. E. Schaepman, and H. Zhao, "Spectral super-resolution reflectance retrieval from remotely sensed imaging spectrometer data," *Opt. Exp.*, vol. 24, no. 17, p. 19905, Aug. 2016, doi: 10.1364/oe.24.019905.

[42] Y. J. Kaufman, D. Tanré, L. A. Remer, E. F. Vermote, A. Chu, and B. N. Holben, "Operational remote sensing of tropospheric aerosol over land from EOS Moderate Resolution Imaging Spectroradiometer," *J. Geophys. Res., Atmos.*, vol. 102, no. D14, pp. 17051–17067, Jul. 1997, doi: 10.1029/96jd03988.

[43] T. Tadono et al., "Generation of the 30 M-mesh global digital surface model by alos prism," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 4, pp. 157–162, Jun. 2016, doi: 10.5194/isprs-archives-xli-b4-157-2016.

[44] A. M. Lingua, P. Boccardo, E. Borgogno-Mondino, and F. G. Tonolo, "Orthorectification of high resolution satellite images," in *Proc. ISPRS Congr.*, Istanbul, Turkey, Jul. 2004, pp. 30–35. [Online]. Available: https://iris.unito.it/bitstream/2318/103885/3/2004_ISPRS.pdf

[45] C. Tao and Y. Hu, "A comprehensive study of the rational function model for photogrammetric processing," *Photogramm. Eng. Remote Sens.*, vol. 67, no. 12, pp. 1347–1357, Dec. 2001. [Online]. Available: https://www.asprs.org/wp-content/uploads/pers/2001journal/december/2001_dec_1347-1357.pdf

[46] S. I. Granshaw, "Bundle adjustment methods in engineering photogrammetry," *Photogrammetric Rec.*, vol. 10, no. 56, pp. 181–207, Oct. 1980, doi: 10.1111/j.1477-9730.1980.tb00020.x.

[47] *Guide To the Expression of Uncertainty in Measurement—Part 1: Introduction*, JCGM GUM-1:2023, Joint Committee Guides Metrology (JCGM), Bureau International des Poids etMesures, Sèvres, France,Jan. 2023, doi: 10.59161/jcgmgum-1-2023.

[48] Q. T. Team. (2010). *A Quality Assurance Framework for Earth Obser-Vation: Principles*. [Online]. Available: https://qa4eo.org/docs/QA4EO_Principles_v4.0.pdf

[49] *Evaluation of Measurement Data–Supplement 1 To the 'Guide To the Expression of Uncertainty in Measurement': Propagation of Distributions Using a Monte Carlo Method*, 1st ed., JCGM, Sèvres, France, 2008. [Online]. Available: https://www.bipm.org/documents/20126/2071204/JCGM_101_2008_E.pdf

[50] J. Fahy et al. (2022). *General Guidance on a Metrological Approach To Fundamental Data Records (FDR), Thematic Data Products (TDPs) and Fiducial Reference Measurements (FRMs)-Metrology Theoreti-cal Basis*. [Online]. Available: https://qa4eo.org/docs/3_Process_Document.pdf

[51] J. Gorroño, L. Guanter, L. Valentin Graf, and F. Gascon, "A framework for the estimation of uncertainties and spectral error correlation in Sentinel-2 level-2A data products," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Jul. 2024, Art. no. 5634613, doi: 10.1109/TGRS.2024.3435021.

[52] P. G. Bohn, "ISOFIT 3.0: Global imaging spectroscopy sur-face and atmospheric retrieval algorithms with an open-source code base," in *Proc. AGU Fall Meeting Abstr.*, Dec. 2023, no. 682, pp. 1–82. [Online]. Available: https://ui.adsabs.harvard.edu/abs/2023AGUFMGC51G0682E/abstract

[53] P. M. Teillet, B. Guindon, and D. G. Goodenough, "On the slope-aspect correction of multispectral scanner data," *Can. J. Remote Sens.*, vol. 8, no. 2, pp. 84–106, Dec. 1982, doi: 10.1080/07038992.1982.10855028.

[54] S. A. Soenen, D. R. Peddle, and C. A. Coburn, "SCS+C: A modified sun-canopy-sensor topographic correction in forested terrain," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 9, pp. 2148–2159, Sep. 2005, doi: 10.1109/TGRS.2005.852480.

[55] J. AL-Doski, S. B. Mansor, H. Paik San, and Z. Khuzaimah, "SCS$C topographic correction to enhance SVM classification accuracy," *J. Eng. Technol. Appl. Phys.*, vol. 2, pp. 32–37, Nov. 2020, doi: 10.33093/jetap.2020.x1.7.

[56] H. N. Salsabila and P. Danoedoro, "The effect of atmospheric and topographic corrections on the vegetation density mapping us-ing several vegetation indices: A case study in Arjuno-Welirang Volcanoes, East Java," *Proc. SPIE*, vol. 12082, pp. 279–284, Dec. 2021, doi: 10.1117/12.2621837.

[57] S. Vanonckelen, S. Lhermitte, and A. Van Rompaey, "The effect of atmospheric and topographic correction methods on land cover classification accuracy," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 24, pp. 9–21, Oct. 2013, doi: 10.1016/j.jag.2013.02.003.

[58] J. Qi et al., "LESS: LargE-scale remote sensing data and image simulation framework over heterogeneous 3D scenes," *Remote Sens. Environ.*, vol. 221, pp. 695–706, Feb. 2019, doi: 10.1016/j.rse.2018.11.036.

[59] Y. Ma, T. He, A. Li, and S. Li, "Evaluation and intercomparison of topographic correction methods based on Landsat images and simulated data," *Remote Sens.*, vol. 13, no. 20, p. 4120, Oct. 2021, doi: 10.3390/rs13204120.

[60] Y. Wang, N. Lauret, and J.-P. Gastellu-Etchegorry, "DART radiative transfer modelling for sloping landscapes," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111902, doi: 10.1016/j.rse.2020.111902.

[61] T. Tadono, J. Takaku, K. Tsutsui, F. Oda, and H. Nagai, "Status of 'ALOS world 3D (AW3D)' global DSM generation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 3822–3825, doi: 10.1109/IGARSS.2015.7326657.

[62] W. Fan et al., "Topographic correction of forest image data based on the canopy reflectance model for sloping terrains in multiple forward mode," *Remote Sens.*, vol. 10, no. 5, p. 717, May 2018, doi: 10.3390/rs10050717.

[63] J. Benesty, J. Chen, Y. Huang, and I. Cohen, "Pearson corre-lation coefficient," in *Noise Reduction in Speech Processing* (Springer Topics Signal Process). Berlin, Germany: Springer, 2009, pp. 1–4, doi: 10.1007/978-3-642-00296-0_5.

[64] M. T. Alexander, D. C. Montgomery, and G. Runger, "Applied statistics and probability for engineers," *Technometrics*, vol. 37, no. 4, p. 455, Nov. 1995, doi: 10.2307/1269738.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.

[65] J. Mittaz, C. J. Merchant, and E. R. Woolliams, "Applying principles of metrology to historical Earth observations from satellites," *Metrologia*, vol. 56, no. 3, May 2019, Art. no. 032002, doi: 10.1088/1681-7575/ab1705.
[66] W. Mi, Q. Kailing, C. Yufeng, Z. Ying, and G. Beibei, "Geometric calibration for GF-5 AHSI camera," *Nat. Remote Sens. Bull.*, vol. 24, no. 4, pp. 345–351, Jan. 2020, doi: 10.11834/jrs.20209198.
[67] N. Origo, J. Gorroño, J. Ryder, J. Nightingale, and A. Bialek, "Fiducial reference measurements for validation of Sentinel-2 and proba-V surface reflectance products," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111690, doi: 10.1016/j.rse.2020.111690.
[68] D. R. Thompson et al., "Retrieval of atmospheric parameters and surface reflectance from visible and shortwave infrared imaging spectroscopy data," *Surveys Geophys.*, vol. 40, no. 3, pp. 333–360, Jul. 2018, doi: 10.1007/s10712-018-9488-9.
[69] D. R. Thompson et al., "Quantifying uncertainty for remote spectroscopy of surface composition," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111898, doi: 10.1016/j.rse.2020.111898.
[70] T. Storch et al., "The EnMAP imaging spectroscopy mission towards operations," *Remote Sens. Environ.*, vol. 294, Aug. 2023, Art. no. 113632, doi: 10.1016/j.rse.2023.113632.
[71] European Space Agency.(2021). *OpenTopography–Copernicus Global Digital Elevation Models*. [Online]. Available: https://portal.opentopography.org/datasetMetadata?otCollectionID=OT.032021.4326.1
[72] M. Abrams, Y. Yamaguchi, and R. Crippen, "Aster global dem (GDEM) version 3," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 4, pp. 593–598, Jun. 2022, doi: 10.5194/isprs-archives-xliii-b4-2022-593-2022.
[73] E. Fahrland et al. (2022). *Public Document Copernicus DEM Co-Pernicus Digital Elevation Model Product Handbook*. [Online]. Available: https://dataspace.copernicus.eu/sites/default/files/media/files/2024-06/geo1988-copernicusdem-spe-002_producthandbook_i5.0.pdf
[74] M. Abrams and R. Crippen. (2021). *ASTER GDEM V3 (ASTER Global DEM) User Guide*. [Online]. Available: https://lpdaac.usgs.gov/documents/434/ASTGTM User Guide_V3.pdf
[75] F. Li et al., "A physics-based atmospheric and BRDF correction for Landsat data over mountainous terrain," *Remote Sens. Environ.*, vol. 124, pp. 756–770, Sep. 2012, doi: 10.1016/j.rse.2012.06.018.
[76] J. Wen et al., "Characterizing land surface anisotropic reflectance over rugged terrain: A review of concepts and recent developments," *Remote Sens.*, vol. 10, no. 3, p. 370, Feb. 2018, doi: 10.3390/rs10030370.
[77] A. Ding et al., "Evaluating the performance of the enhanced ross-li models in characterizing BRDF/Albedo/NBAR characteristics for various land cover types in the POLDER database," *Remote Sens.*, vol. 16, no. 12, p. 2119, Jun. 2024, doi: 10.3390/rs16122119.
[78] A. Lyapustin et al., "Multiangle implementation of atmospheric correction (MAIAC): 2. Aerosol algorithm," *J. Geophys. Res.*, vol. 116, no. D3, pp. D03211-1–D03211-15, Feb. 2011, doi: 10.1029/2010jd014986.
[79] E. Emili et al., "High spatial resolution aerosol retrieval with MAIAC: Application to mountain regions," *J. Geophys. Research: Atmos.*, vol. 116, no. D23, pp. 1–12, Dec. 2011, doi: 10.1029/2011jd016297.
[80] D. R. Thompson et al., "Optimal estimation of spectral surface reflectance in challenging atmospheres," *Remote Sens. Environ.*, vol. 232, Oct. 2019, Art. no. 111258, doi: 10.1016/j.rse.2019.111258.

Guorui Jia (Senior Member, IEEE) received the B.S. and Ph.D. degrees from Beihang University, Beijing, China, in 2005 and 2011, respectively.

He was a Visiting Scholar with the University of Zurich, Zürich, Switzerland, from 2015 to 2016. He is currently an Associate Professor with the School of Instrumentation and Optoelectronic Engineering, Beihang University. His research interests include remote sensing modeling, hyperspectral instrument modeling, hyperspectral data simulation, and preprocessing.

Yujie Zhao received the B.S. degree from Beihang University, Beijing, China, in 2021, where he is currently pursuing the Ph.D. degree in precision instruments and machinery with the School of Instrumentation and Optoelectronic Engineering.

His research interests include hyperspectral data preprocessing and uncertainty quantification of hyperspectral data.

Shuhan Liu received the master's degree from the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing, China, in 2014.

Her research interests include remote sensing modeling and hyperspectral data preprocessing.

Chenchao Xiao received the B.S. degree from the School of Geographical Sciences, Northwest University, Kirkland, WA, USA, in 2007, and the Ph.D. degree in cartography and geographic information systems from Peking University, Beijing, China, in 2017.

His research interests include remote sensing data processing and data analysis.

Huijie Zhao received the B.S., M.S., and Ph.D. degrees from Harbin Institute of Technology, Harbin, China, in 1988, 1991, and 1994, respectively.

She joined Beihang University, Beijing, China, in 1995. From 1999 to 2000, she was a Post-Doctoral Fellow with the University of Michigan, Ann Arbor, MI, USA. She is currently a Professor with the School of Artificial Intelligence, Beihang University. She is the Director of the Key Laboratory of Aerospace Broad Band Detection and Intelligent Perception, Ministry of Industry and Information Technology, Beijing, and the Deputy Director of the Key Laboratory of Precision Opto-Mechatronics Technology, Ministry of Education, Beijing. Her research interests include hyperspectral remote sensing instruments and data processing, machine vision, and 3-D inspection.

Dr. Zhao was awarded the Excellent Teacher in Beijing in 1997 and the Outstanding Postgraduate Supervisor in Beijing in 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:09:34 UTC from IEEE Xplore. Restrictions apply.